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RESEARCH ARTICLE

# Lampricide 3-trifluoromethyl-4-nitrophenol (TFM) impacts whole-organism oxygen consumption in non-target fishes

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## ABSTRACT

The application of the lampricide 3-trifluoromethyl-4-nitrophenol (TFM) to streams infested with larval sea lamprey constitutes the backbone of the Sea Lamprey Control Program in the Laurentian Great Lakes, but native species have greatly varying sensitivities to TFM. Earlier work by D'Souza and colleagues revealed that the whole-animal oxygen consumption of larval sea lamprey responded in a dose-dependent manner to increasing TFM concentrations. We set out to determine (1) if TFM exposure leads to stepwise increases in whole-animal oxygen consumption in non-target fishes and (2) if exposure to lethal TFM concentrations leads to increases in oxygen consumption comparable to the animals' maximum metabolic rate (MMR). Using intermittent-flow respirometry, we tested three non-target fish species known to differ in their sensitivity to TFM: young-of-the-year lake sturgeon (sensitive), juvenile rainbow trout (intermediate), and juvenile smallmouth bass (resistant). We confirmed that all three species exhibit increases in oxygen consumption during exposure to increasing concentrations of TFM but found that the relationship between the MMR and oxygen consumption prior to death during lethal TFM exposure is complex and species dependent. While this complex relationship broadens the need for species-specific testing, it opens up interesting possibilities for both (1) *in situ* testing, e.g. through the deployment of riverside field respirometry systems to monitor real-time responses of both target and non-target species to TFM treatments, and (2) *ex situ* testing, e.g. through the establishment of less-lethal assays to determine TFM sensitivity under various species×temperature×water chemistry combinations.

**KEY WORDS: Comparative physiology, Respirometry, Non-target, Great Lakes, Pesticide, Oxygen consumption**

## INTRODUCTION

Spanning more than 240,000 km$^2$, the contiguous Laurentian Great Lakes of North America (henceforth referred to as the Great Lakes) border both Canada and the US and represent the largest system of superficial fresh water on Earth. The Great Lakes are of immense cultural and economic importance to lakeshore communities and Indigenous peoples in the region, including the Métis, First Nations, and Tribal Nations (Pirie et al., 2025; Smith and Tibbles, 1980), and support a vast recreational and commercial fisheries industry.

The accidental introduction of sea lamprey (*Petromyzon marinus*) in the 19th century destabilized the Great Lakes ecosystem and economy (Applegate, 1951). Sea lamprey gained access to the Great Lakes from the Atlantic Ocean through a series of anthropogenic canals (Applegate, 1951; Eshenroder, 2014; Gaden et al., 2021). This jawless fish tends to target larger species of fish in the Great Lakes that are important for both commercial and recreational fisheries, thereby threatening Great Lakes economies (Gaden et al., 2021; Smith and Tibbles, 1980). The growing severity of the sea lamprey invasion led to the formation of the Great Lakes Fishery Commission (GLFC) in 1955, with a mandate to eradicate sea lamprey from the Great Lakes. The application of lampricide 3-trifluoromethyl-4-nitrophenol (TFM) to streams infested with larval sea lamprey now constitutes the backbone of the control program, as the sedentary nature of larval sea lamprey make them an ideal life stage for targeted control (Siefkes, 2017; Sullivan et al., 2021). The application of lampricides is particularly successful both because it regularly ensures the elimination of over 90% of the larval population in a stream and because it eliminates the sea lamprey before they reach the parasitic life stage, thereby avoiding parasitic damage to native fish populations (Applegate, 1951; Siefkes, 2017; Sullivan et al., 2021).

TFM is characterized as a protonophore due to its ability to alter the mitochondrial membrane's permeability to protons and dismantle the proton motive force used to produce ATP (Birceanu et al., 2009, 2011; Kawatski et al., 1974; Viant et al., 2001). The resulting decreased intermembrane proton concentration not only lowers ATP production but also triggers an increase in proton pumping by the electron transport chain (ETC) complexes in an attempt to restore the proton motive force (Borowiec et al., 2022; Huerta et al., 2020; Wilkie et al., 2019). Because oxygen is the final electron acceptor of the ETC, the increased proton pumping observed during TFM exposure leads to an increased mitochondrial oxygen consumption (Birceanu et al., 2011; Borowiec et al., 2022; Huerta et al., 2020). Recently, D'Souza et al. (2025) found that this increase in cellular oxygen consumption during TFM exposure in larval sea lamprey is reflected in their whole-animal oxygen consumption and can be measured using whole-animal intermittent-flow respirometry.

Sea lamprey are particularly vulnerable to the effects of TFM because they have a weak ability to biotransform and detoxify it using phase II biotransformation pathways in the liver (Bussy et al., 2018; Kane et al., 1994; Lech and Statham, 1975). However, non-target Great Lakes species vary greatly in their sensitivity to TFM. For example, previous research on TFM sensitivity has identified

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the Centrarchidae, such as the smallmouth bass, as very TFM tolerant (Boogaard et al., 2003). Salmonids have a moderate sensitivity to TFM, with studies using rainbow trout finding they can detoxify TFM through biotransformation and excretion of the molecule in the process of glucuronidation (Kane et al., 1994; Lech and Costrini, 1972; Lech and Statham, 1975). On the other hand, the threatened lake sturgeon (*Acipenser fulvescens*) is highly sensitive to TFM when it is less than 100 mm in length, with 50% lethal concentrations (LC50) similar to those of larval sea lamprey (Boogaard et al., 2003). Furthermore, tolerance to TFM is also dependent on environmental factors such as temperature, pH, and alkalinity (Bills et al., 2003; Hunn and Allen, 1974; McDonald and Kolar, 2007), which presents challenges toward accurately predicting dangerous TFM concentrations for non-target species.

Improving our understanding of the sensitivity of native Great Lakes species to TFM fits under a primary focus of the GLFC's sea lamprey control program. However, traditional TFM sensitivity experiments necessitate lethal endpoints for study animals (e.g. Applegate and King, 1962; Bills et al., 2003; Boogaard et al., 2003). It is not yet clear whether the relationship between whole-animal oxygen consumption and TFM exposure reported for larval sea lamprey by D'Souza et al. (2025) can be observed in non-target species. D'Souza et al. (2025) also postulated that the oxygen consumption of larval sea lamprey at lethal TFM concentrations (before mortality) is comparable to their maximum metabolic rate (MMR), which sets forth a question of whether lethal TFM sensitivity can be derived from a species' MMR.

In this study, we aimed to determine (1) if TFM exposure leads to stepwise increases in whole-animal oxygen consumption in non-target species and (2) if exposure to lethal TFM concentrations leads to increases in oxygen consumption comparable to the animals' MMR. We hypothesize that the increases in mitochondrial oxygen consumption induced by TFM exposure will lead to an increase in oxygen consumption at the whole-body level in non-target species just as is observed in larval sea lamprey and that during exposure to lethal concentrations of TFM, non-target Great Lakes fish species will elevate their oxygen consumption to levels that are not significantly different from their MMR. We tested these hypotheses using intermittent-flow respirometry to measure the mass-specific oxygen consumption ($\dot{M}_{O_2}$) of three non-target species exposed to a range of sublethal and lethal concentrations of TFM: young-of-the-year lake sturgeon, juvenile rainbow trout, and juvenile smallmouth bass. These three species were selected to encompass a wide range of TFM sensitivity levels, with young-of-the-year lake sturgeon being a species of high conservation interest that is highly sensitive during early life stages, rainbow trout being a moderately resistant species that is well studied and serves as a comparable control, and juvenile smallmouth bass being a highly resistant species that serves as an interesting comparison to more sensitive models (Boogaard et al., 2003).

## RESULTS

A total of 36 trials (11 chasing trials and 25 TFM exposure trials) were conducted between 9 November 2025 and 24 February 2026 (~3.5 months). A detailed list of the trials, species tested per trial, nominal and observed TFM concentrations, and the averages and 95% range of values recorded for pH, alkalinity, and TFM is provided in Table S1. Summarily, average observed TFM concentrations remained within 0.2 mg $l^{-1}$ of the nominal TFM concentrations, with little to no variation throughout the 12-h exposure period. Alkalinity averaged 182 mg $l^{-1}$ $CaCO_3$ during the TFM trials, and 143 mg $l^{-1}$ $CaCO_3$ in the chasing trials, mainly due to lower alkalinity at the initial chasing trials. The alkalinity values for four trials (three TFM and one chasing) were lost and could not be recovered but were known to be within expected targets as the trials were performed. pH generally averaged between 8.4 and 8.6. Further run-by-run details are provided in Table S1.

A summary description of all the metrics calculated in this study is provided in Table 1. Unless stated otherwise, effect sizes are described for hypothetical average animals exposed to control conditions (i.e. 0 TFM). The statistical results of all the generalized linear models (GLMs) and generalized additive models (GAMs) conducted to explore the impacts of TFM and covariates on the various metrics calculated are provided in Table 2 (lake sturgeon models), Table 3 (rainbow trout models), and Table 4 (smallmouth bass models), with the written results highlighting the relevant findings. The statistical results of secondary tests conducted to compare two or more calculated metrics (e.g. $MMR_S$ versus $MMR_F$, or $MMR_F$ versus $\dot{M}_{O_2\,TFM\,q0.8}$ and $\dot{M}_{O_2\,TFM\,Peak}$) are described directly in the text.

### Standard metabolic rate (SMR) remained relatively stable throughout the experiments

Pre-chase SMR averaged 2.3±0.1 µmol $O_2$ $g^{-1}$ $h^{-1}$ for lake sturgeon ($n$=45, average $\dot{M}_{O_2}$ values per animal=403), 5.1±0.1 µmol $O_2$ $g^{-1}$ $h^{-1}$ for rainbow trout ($n$=18, average $\dot{M}_{O_2}$ values per animal=412), and 2.6±0.1 µmol $O_2$ $g^{-1}$ $h^{-1}$ for smallmouth bass ($n$=14, average $\dot{M}_{O_2}$ values per animal=447). The recorded pre-chase SMR values for lake sturgeon lowered marginally but significantly as a function of day since the first experiment (−0.014 µmol $O_2$ $g^{-1}$ $h^{-1}$ per day; ~1% of the recorded within-day variation, Table 2). Lake sturgeon chasing trials clustered on days 0-10 (average SMR 2.7 µmol $O_2$ $g^{-1}$ $h^{-1}$) and days 70-77 (average SMR 1.9 µmol $O_2$ $g^{-1}$ $h^{-1}$), with the animals on the second group being larger (average 3.0 and 4.7 g for the first

**Table 1. Calculated metrics and their respective descriptions**

| Variable | Description |
|---|---|
| $SMR_{Chase}$ | 20th percentile of the $\dot{M}_{O_2}$ values obtained prior to chasing in the chasing trials |
| $MMR_S$ | Highest spontaneous $\dot{M}_{O_2}$ measurement, prior to chasing |
| $MMR_F$ | Highest $\dot{M}_{O_2}$ measurement following forced, exhaustive exercise |
| AS | Aerobic scope; $MMR_F$ minus the animal's respective $SMR_{Chase}$ |
| CE | Chasing efficacy (a measure of the relationship between $MMR_F$ and $MMR_S$) |
| $EPOC_{Chase}$ | Excess post-exercise oxygen consumption (chasing trials) |
| $t_{1/2\ Chase}$ | Half-time of oxygen consumption recovery to SMR levels following chasing |
| $SMR_{TFM}$ | 20th percentile of the $\dot{M}_{O_2}$ values obtained prior to TFM exposure in the toxicity trials |
| $\dot{M}_{O_2\,TFM\,q0.8}$ | 80th percentile of the $\dot{M}_{O_2}$ values obtained during TFM exposure |
| $\dot{M}_{O_2\,TFM\,Peak}$ | Highest $\dot{M}_{O_2}$ measurement obtained during TFM exposure |
| $\Delta\dot{M}_{O_2\,TFM\,q0.8}$ | $\dot{M}_{O_2\,TFM\,q0.8}$ minus the animal's respective $SMR_{TFM}$ |
| $\Delta\dot{M}_{O_2\,TFM\,Peak}$ | $\dot{M}_{O_2\,TFM\,Peak}$ minus the animal's respective $SMR_{TFM}$ |
| $EPOC_{TFM}$ | Excess post-exposure oxygen consumption (toxicity trials) |
| $t_{1/2\ TFM}$ | Half-time of oxygen consumption recovery to SMR levels following TFM exposure |

Metrics obtained from the chasing trials are listed first, followed by the metrics obtained from the TFM exposure trials.

**Table 2. Statistics summary of the models performed on the variables calculated for lake sturgeon, including the intercept (α) and slopes (β) for each response variable**

| Lake sturgeon | | | | TFM (mg/l) | | | Day | | | Background (% of SMR) | | | Mass (g) | | | Condition factor (k) | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Variable | Model | *n* | α | β | F | *P* | β | F | *P* | β | F | *P* | β | F | *P* | β | F | *P* |
| $SMR_{Chase}$ | Gamma GLM | 45 | 0.875 | - | - | - | **−0.006** | **19.9** | **<0.001** | 0.002 | 0.24 | 0.629 | 0.047 | 1.43 | 0.239 | −0.133 | 0.04 | 0.841 |
| $MMR_F$ | Gamma GLM | 45 | 2.386 | - | - | - | **−0.002** | **6.11** | **0.018** | −0.003 | 0.84 | 0.365 | 0.048 | 2.63 | 0.113 | −0.872 | 3.12 | 0.085 |
| AS | Gamma GLM | 45 | 2.155 | - | - | - | −0.001 | 0.82 | 0.372 | −0.006 | 2.11 | 0.154 | 0.045 | 1.66 | 0.205 | −1.134 | 3.68 | 0.062 |
| $SMR_{TFM}$ | Gamma GLM | 66 | 1.08 | **0.026** | **7.48** | **0.008** | **−0.005** | **24.8** | **<0.001** | **−0.007** | **9.98** | **0.002** | 0.000 | 0.00 | 0.996 | 0.226 | 0.26 | 0.610 |
| $\dot{M}_{O_2TFM\,q0.8}$ | Gaussian GLM | 66 | 5.119 | **1.094** | **504** | **<0.001** | **−0.021** | **14.0** | **<0.001** | −0.01 | 0.82 | 0.37 | −0.141 | 0.94 | 0.337 | −4.005 | 2.89 | 0.095 |
| $\dot{M}_{O_2TFM\,Peak}$ | Gaussian GLM | 66 | 10.70 | **0.675** | **113** | **<0.001** | **−0.018** | **5.77** | **0.019** | −0.014 | 0.91 | 0.343 | −0.161 | 0.71 | 0.402 | **−8.227** | **7.15** | **0.010** |
| $\Delta\dot{M}_{O_2TFM\,q0.8}$ | Gaussian GLM | 66 | 2.261 | **1.036** | **605** | **<0.001** | −0.009 | 3.44 | 0.069 | 0.002 | 0.06 | 0.802 | −0.146 | 1.35 | 0.25 | **−4.387** | **4.64** | **0.035** |
| $\Delta\dot{M}_{O_2TFM\,Peak}$ | Gaussian GLM | 66 | 7.843 | **0.617** | **121** | **<0.001** | 0.006 | 0.75 | 0.390 | −0.001 | 0.01 | 0.913 | −0.166 | 0.97 | 0.328 | **−8.609** | **10.0** | **0.002** |
| $EPOC_{TFM}$ | Gamma GLM | 44 | 1.488 | 0.059 | 3.67 | 0.063 | **0.004** | **4.56** | **0.039** | 0.004 | 0.75 | 0.392 | −0.010 | 0.04 | 0.848 | −0.045 | 0.00 | 0.958 |
| $t_{1/2\ TFM}$ (4-8 mg $l^{-1}$) | Gamma GLM | 38 | 0.875 | 0.009 | 0.01 | 0.916 | 0.004 | 1.66 | 0.207 | 0 | 0 | 0.988 | −0.093 | 1.63 | 0.211 | 0.441 | 0.14 | 0.713 |

Significant variables ($P$<0.05) are highlighted in bold. Gamma models were run with log link, and Gaussian models were run with identity link. For more details on the definitions of the variables, please see Table 1.

**Table 3. Statistics summary of the GLMs performed on the variables calculated for rainbow trout, including the intercept (α) and slopes (β) for each response variable**

| Rainbow trout | | | | TFM (mg/l) | | | Day | | | Background (% of SMR) | | | Mass (g) | | | Condition factor (k) | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Variable | Model | *n* | α | β | F | *P* | β | F | *P* | β | F | *P* | β | F | *P* | β | F | *P* |
| $SMR_{Chase}$ | Gamma GLM | 18 | 1.697 | - | - | - | 0.03 | 4.3 | 0.059 | −0.001 | 0.27 | 0.609 | −0.003 | 0.08 | 0.787 | −0.069 | 0.12 | 0.73 |
| $MMR_F$ | Gamma GLM | 18 | 3.299 | - | - | - | −0.036 | 3.48 | 0.085 | −0.003 | 0.84 | 0.375 | −0.009 | 0.32 | 0.582 | 0.096 | 0.13 | 0.72 |
| AS | Gamma GLM | 18 | 3.07 | - | - | - | −0.053 | 4.19 | 0.061 | −0.003 | 0.59 | 0.457 | −0.01 | 0.21 | 0.655 | 0.131 | 0.14 | 0.71 |
| $SMR_{TFM}$ | Gamma GLM | 77 | 2.259 | 0.001 | 0.66 | 0.421 | −0.002 | 1.17 | 0.284 | **−0.014** | **4.02** | **0.049** | **−0.012** | **14.4** | **< 0.001** | **−0.513** | **10.1** | **0.002** |
| $\dot{M}_{O_2TFM\,q0.8}$ | Gamma GLM | 77 | 2.248 | **0.011** | **13.3** | **0.001** | −0.006 | 2.75 | 0.102 | 0.017 | 1.5 | 0.225 | −0.002 | 0.12 | 0.734 | −0.001 | 0 | 0.998 |
| $\dot{M}_{O_2TFM\,Peak}$ | Gamma GLM | 77 | 3.487 | −0.003 | 1.36 | 0.248 | −0.001 | 0.1 | 0.755 | 0.004 | 0.14 | 0.708 | −0.003 | 0.4 | 0.529 | −0.45 | 3.04 | 0.086 |
| $\Delta\dot{M}_{O_2TFM\,q0.8}$ | Gaussian GLM | 77 | 1.94 | **0.109** | **14.1** | **<0.001** | −0.051 | 1.92 | 0.17 | 0.259 | 3.45 | 0.068 | 0.012 | 0.03 | 0.855 | 2.254 | 0.48 | 0.489 |
| $\Delta\dot{M}_{O_2TFM\,Peak}$ | Gaussian GLM | 77 | 20.75 | −0.048 | 1.76 | 0.189 | −0.01 | 0.05 | 0.823 | 0.111 | 0.41 | 0.526 | −0.011 | 0.02 | 0.891 | −5.485 | 1.83 | 0.181 |
| $EPOC_{TFM}$ | Gamma GAM | 48 | 2.079 | NL | 1.49 | 0.206 | −0.002 | 0.17 | 0.683 | −0.223 | 0.41 | 0.528 | 0.005 | 0.25 | 0.618 | 0.011 | 0 | 0.989 |
| $t_{1/2\ TFM}$ | Gamma GAM | 46 | −1.106 | **NL** | **4.25** | **0.015** | 0.005 | 0.31 | 0.583 | −0.076 | 2.09 | 0.157 | −0.004 | 0.08 | 0.773 | 1.909 | 2.32 | 0.136 |

Significant variables ($P$<0.05) are highlighted in bold. Gamma models were run with log link, and Gaussian models were run with identity link. For more details on the definitions of the variables, please see Table 1. NL=non-linear.





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**Table 4. Statistics summary of the GLMs performed on the variables calculated for smallmouth bass, including the intercept (α) and slopes (β) for each response variable**

| Smallmouth bass | | | | TFM (mg/l) | | | Day | | | Background (% of SMR) | | | Mass (g) | | | Condition factor (k) | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Variable | Model | *n* | α | β | F | *P* | β | F | *P* | β | F | *P* | β | F | *P* | β | F | *P* |
| $SMR_{Chase}$ | Gamma GLM | 14 | 1.078 | - | - | - | 0.05 | 4.80 | 0.056 | 0.001 | 0.67 | 0.435 | −0.038 | 4.76 | 0.057 | −0.065 | 0.06 | 0.812 |
| $MMR_F$ | Gamma GLM | 14 | 1.869 | - | - | - | 0.017 | 0.88 | 0.373 | 0.001 | 0.12 | 0.736 | **0.05** | **12.3** | **0.007** | 0.251 | 1.34 | 0.277 |
| AS | Gamma GLM | 14 | 1.412 | - | - | - | 0.009 | 0.16 | 0.700 | 0 | 0.02 | 0.897 | **0.075** | **20.1** | **0.002** | 0.338 | 1.79 | 0.213 |
| $SMR_{TFM}$ | Gamma GLM | 59 | 2.143 | 0 | 0.04 | 0.850 | 0 | 0.05 | 0.818 | −0.002 | 0.51 | 0.480 | **−0.06** | **32.3** | **<0.001** | **−0.641** | **56.5** | **<0.001** |
| $\dot{M}_{O_2,TFM\,q0.8}$ | Gamma GLM | 59 | 2.346 | **0.014** | **63.6** | **<0.001** | 0.002 | 0.86 | 0.357 | −0.004 | 1.42 | 0.238 | **−0.054** | **13.2** | **0.001** | **−0.470** | **15.3** | **<0.001** |
| $\dot{M}_{O_2,TFM\,Peak}$ | Gamma GLM | 59 | 3.365 | 0.001 | 0.15 | 0.700 | 0.006 | 1.46 | 0.232 | −0.007 | 1.10 | 0.299 | **−0.082** | **8.3** | **0.006** | **−0.503** | **5.36** | **0.024** |
| $\Delta\dot{M}_{O_2,TFM\,q0.8}$ | Gaussian GLM | 59 | 3.482 | **0.101** | **83.3** | **<0.001** | 0.013 | 0.79 | 0.379 | −0.030 | 2.17 | 0.147 | **−0.209** | **5.21** | **0.027** | −0.968 | 1.69 | 0.199 |
| $\Delta\dot{M}_{O_2,TFM\,Peak}$ | Gaussian GLM | 59 | 16.70 | 0.003 | 0.01 | 0.931 | 0.054 | 1.04 | 0.313 | −0.045 | 0.39 | 0.534 | −0.639 | 3.97 | 0.052 | −4.597 | 3.10 | 0.084 |
| $EPOC_{TFM}$ | Gamma GLM | 45 | 1.617 | 0.01 | 3.19 | 0.082 | **0.024** | **10.8** | **0.002** | −0.005 | 0.36 | 0.551 | −0.024 | 0.39 | 0.538 | −0.581 | 2.19 | 0.147 |
| $t_{1/2\ TFM}$ (24-60 mg $l^{-1}$) | Gamma GLM | 38 | 0.616 | −0.007 | 0.41 | 0.528 | 0.006 | 0.36 | 0.553 | 0.005 | 0.18 | 0.673 | 0.016 | 0.08 | 0.774 | 0.151 | 0.1 | 0.749 |

Significant variables ($P<0.05$) are highlighted in bold. Gamma models were run with log link, and Gaussian models were run with identity link. For more details on the definitions of the variables, please see Table 1.

and second sets of trials, respectively), which could have contributed to the detected difference (although neither mass nor condition was deemed significant). None of the tested variables had a significant impact on the pre-chase SMR of rainbow trout or smallmouth bass.

Pre-exposure SMR averaged 2.4±0.1 µmol $O_2$ $g^{-1}$ $h^{-1}$ for lake sturgeon ($n$=66, average $\dot{M}_{O_2}$ values per animal=238), 3.9±0.1 µmol $O_2$ $g^{-1}$ $h^{-1}$ for rainbow trout ($n$=77, average $\dot{M}_{O_2}$ values per animal=214), and 2.7±0.1 µmol $O_2$ $g^{-1}$ $h^{-1}$ for smallmouth bass ($n$=59, average $\dot{M}_{O_2}$ values per animal=253). Unexpectedly, the model revealed a significant but small difference in pre-exposure SMR between lake sturgeon exposure groups (0.06 µmol $O_2$ $g^{-1}$ $h^{-1}$ per mg $l^{-1}$ of TFM, Table 2). However, this represents a small portion of the within-group variation in SMR for lake sturgeon (~2.5%), so we do not expect this to have a strong impact in the interpretation of the results. Similarly to the pre-chase SMR estimates, pre-exposure SMR values for lake sturgeon were also found to have lowered marginally but significantly as a function of day since the first experiment (−0.01 µmol $O_2$ $g^{-1}$ $h^{-1}$ per day). Finally, chambers with higher background levels tended to record lower SMR values for lake sturgeon, even after background correction (−0.01 µmol $O_2$ $g^{-1}$ $h^{-1}$ per percentage point increase in background relevance; range 3-73%). The same was also the case for rainbow trout (−0.05 µmol $O_2$ $g^{-1}$ $h^{-1}$ per percentage point increase in background relevance, Table 3). Both rainbow trout and smallmouth bass showed lower SMR with larger mass (−0.04 and −0.14 µmol $O_2$ $g^{-1}$ $h^{-1}$ per gram of increase, respectively) and with higher condition factors (0.20 and 0.17 µmol $O_2$ $g^{-1}$ $h^{-1}$ per 0.1 increase in condition, respectively, Table 4).

### Chasing does not elevate $MMR_F$ beyond $MMR_S$ for lake sturgeon and smallmouth bass

$MMR_F$ averaged 8.6±0.2 µmol $O_2$ $g^{-1}$ $h^{-1}$ for lake sturgeon ($n$=45), 23.7±0.5 µmol $O_2$ $g^{-1}$ $h^{-1}$ for rainbow trout ($n$=18), and 11.5±0.4 µmol $O_2$ $g^{-1}$ $h^{-1}$ for smallmouth bass ($n$=14). Aerobic scope (AS) averaged 6.3±0.2 µmol $O_2$ $g^{-1}$ $h^{-1}$ for lake sturgeon ($n$=45), 18.6±0.6 µmol $O_2$ $g^{-1}$ $h^{-1}$ for rainbow trout ($n$=18), and 8.9±0.4 µmol $O_2$ $g^{-1}$ $h^{-1}$ for smallmouth bass ($n$=14). Finally, the highest spontaneous activity prior to chasing ($MMR_S$) averaged 9.0±0.3 µmol $O_2$ $g^{-1}$ $h^{-1}$ for lake sturgeon ($n$=45), 22.0±0.7 µmol $O_2$ $g^{-1}$ $h^{-1}$ for rainbow trout ($n$=18), and 11.1±0.7 µmol $O_2$ $g^{-1}$ $h^{-1}$ for smallmouth bass ($n$=14). Progression of the $\Delta\dot{M}_{O_2}$ throughout the chasing trials for the three species are shown in Fig. 1. Twenty-five lake sturgeons (56%) showed higher $MMR_S$ than $MMR_F$ (Fig. 2), with chasing efficacy (CE) averaging −1.0±2.9%. Conversely, only five rainbow trout (28%) and five smallmouth bass (36%) showed higher $MMR_S$ than $MMR_F$, with CE averaging 9.6±3.9% and 8.3±5.9%, respectively.

Similarly to the SMR estimates, $MMR_F$ values for lake sturgeon were also found to have lowered marginally but significantly as a function of day since the first experiment (−0.02 µmol $O_2$ $g^{-1}$ $h^{-1}$ per day, ~0.3% of the within-day variation, Table 2), but this was not seen for AS. None of the tested variables significantly affected the $MMR_F$ or AS of rainbow trout. Finally, both the $MMR_F$ and AS of smallmouth bass increased as a function of animal mass (0.5-0.7 µmol $O_2$ $g^{-1}$ $h^{-1}$ per gram of increase for $MMR_F$ and 0.6-0.9 µmol $O_2$ $g^{-1}$ $h^{-1}$ per gram of increase for AS).

CE was not significantly different from 0 for lake sturgeon (one-sample two-sided $t$-test, $t$=−0.35, d.f.=44, $P$=0.72) nor for smallmouth bass (one-sample two-sided $t$-test, $t$=1.5, d.f.=13, $P$=0.18). However, the variation in $MMR_F$ values was significantly

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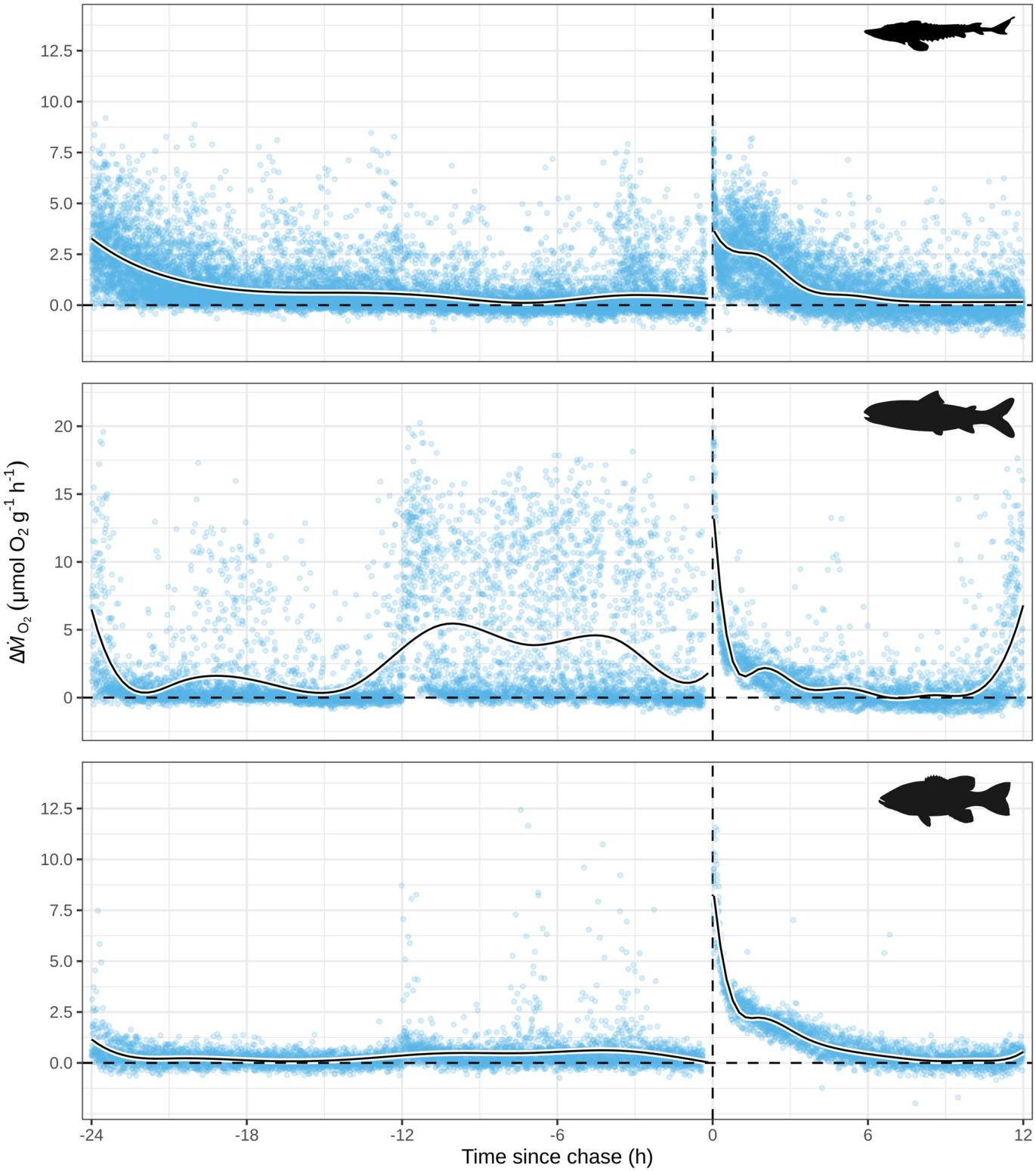


**Fig. 1. Delta oxygen consumption (absolute $\dot{M}_{O_2}$ minus the animal's respective SMR) of lake sturgeon (*n*=45), rainbow trout (*n*=18), and smallmouth bass (*n*=14) throughout 11 chase trials**. The dashed horizontal line at $\Delta\dot{M}_{O_2} = 0$ highlights the SMR state. The vertical dashed line indicates the point where the animals were chased to exhaustion. The solid line shows the generalized additive-modelled population trend over time. Chasing was conducted approximately at the same time of day every day (19:00-20:00) to minimize the effects of circadian rhythms, as those are noticeable for rainbow trout. The MMR values of four rainbow trout were omitted to optimize vertical plot space and ensure comparable Y-axes between Figs 1 and 3.

lower than that in $MMR_S$ values for both lake sturgeon (Fig. 2; two-sided F-test, F=0.56, d.f.=44, *P*=0.011) and smallmouth bass (two-sided F-test, F=0.31, d.f.=13, *P*=0.044). Finally, in rainbow trout, CE was significantly higher than 0 (one-sample two-sided *t*-test, *t*=2.46, d.f.=17, *P*=0.024), but $MMR_F$ and $MMR_S$ values showed similar variation (two-sided F-test, F=0.67, d.f.=17, *P*=0.41).

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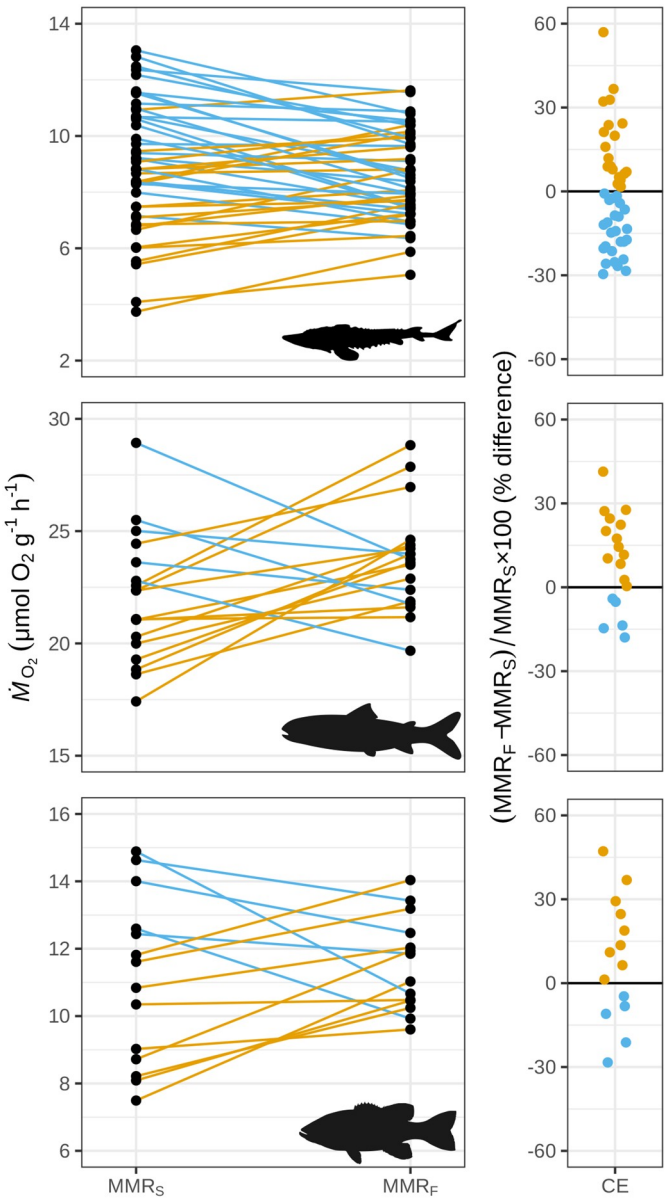


**Fig. 2. Relative magnitude of the oxygen consumption rates induced by spontaneously occurring bursts of activity (MMR$_S$) and by forced chasing (MMR$_F$) for lake sturgeon (*n*=45), rainbow trout (*n*=18), and smallmouth bass (*n*=14). CE is shown on the right-side panels.** Positive CE values (where chasing induced higher $\dot{M}_{O_2}$ than spontaneous activity) are shown in orange, while negative CEs are drawn in blue, with the respective relationships also drawn on the left-side panels.

### TFM exposure consistently impacts $\dot{M}_{O_2TFM\,q0.8}$, but not always $\dot{M}_{O_2TFM\,Peak}$

TFM exposure led to dose-dependent increases in $\dot{M}_{O_2}$ in all three species (Fig. 3). Exposure TFM concentrations ranged from 0 to 8 mg $l^{-1}$ for lake sturgeon and 0 to 60 mg $l^{-1}$ for rainbow trout and smallmouth bass. Notably, 12 lake sturgeons died during exposure to 8 mg $l^{-1}$ of TFM. Furthermore, 16 rainbow trout died during exposure (three, four, three, and six animals at 36, 48, 54, and 60 mg $l^{-1}$ of TFM, respectively). Two smallmouth bass died during exposure (one at 48 mg/l and one at 60 mg $l^{-1}$). The delta variants of the exposure variables (namely, $\Delta\dot{M}_{O_2TFM\,q0.8}$ and $\Delta\dot{M}_{O_2TFM\,Peak}$) followed similar trends to their absolute equivalents ($\dot{M}_{O_2TFM\,q0.8}$ and $\dot{M}_{O_2TFM\,Peak}$). In the interest of brevity, those are only brought up below when they differ from the absolute variables. The full list of statistical tests from GLMs and GAMs is available in Tables 2-4.

In lake sturgeon, exposure to 8 mg $l^{-1}$ of TFM raised oxygen consumption to an average $\dot{M}_{O_2TFM\,q0.8}$ of 10.4±0.5 μmol $O_2$ $g^{-1}$ $h^{-1}$ and an average $\dot{M}_{O_2TFM\,Peak}$ of 11.5±0.5 μmol $O_2$ $g^{-1}$ $h^{-1}$ (*n*=20). In rainbow trout, exposure to 60 mg $l^{-1}$ of TFM raised oxygen consumption to an average $\dot{M}_{O_2TFM\,q0.8}$ of 14.2±0.4 μmol $O_2$ $g^{-1}$ $h^{-1}$ and an average $\dot{M}_{O_2TFM\,Peak}$ of 15.8±0.5 μmol $O_2$ $g^{-1}$ $h^{-1}$ (*n*=6). Finally, in smallmouth bass, exposure to 60 mg $l^{-1}$ of TFM raised oxygen consumption to an average $\dot{M}_{O_2TFM\,q0.8}$ of 9.7±1.2 μmol $O_2$ $g^{-1}$ $h^{-1}$ and an average $\dot{M}_{O_2TFM\,Peak}$ of 11.2±1.7 μmol $O_2$ $g^{-1}$ $h^{-1}$ (*n*=6).

TFM concentration had a strong impact on the $\dot{M}_{O_2TFM\,q0.8}$ and $\dot{M}_{O_2TFM\,Peak}$ of lake sturgeon, with both variables increasing linearly with exposure to increasing TFM concentrations (1.1 and 0.7 μmol $O_2$ $g^{-1}$ $h^{-1}$ mg $l^{-1}$ of TFM, respectively; Fig. 4; Table 2). Experimental day had a significant negative effect on $\dot{M}_{O_2TFM\,q0.8}$ of lake sturgeon (−0.02 μmol $O_2$ $g^{-1}$ $h^{-1}$ per day). However, this was not the case for $\Delta\dot{M}_{O_2TFM\,q0.8}$.

Interestingly, while the $\dot{M}_{O_2TFM\,q0.8}$ of rainbow trout increased with increasing TFM concentrations (0.09-0.17 μmol $O_2$ $g^{-1}$ $h^{-1}$ per mg $l^{-1}$ of TFM; Fig. 4; Table 3), that was not the case for $\dot{M}_{O_2TFM\,Peak}$. This is likely the result of high, naturally occurring peaks of activity regardless of TFM concentration. None of the other tested variables impacted the absolute or delta variables in rainbow trout during the TFM exposure period.

In smallmouth bass, similarly to rainbow trout, exposure to higher TFM concentrations led to stepwise increases in $\dot{M}_{O_2TFM\,q0.8}$ (0.06-0.14 μmol $O_2$ $g^{-1}$ $h^{-1}$ per mg $l^{-1}$ of TFM; Fig. 4; Table 4), but not $\dot{M}_{O_2TFM\,Peak}$. Larger smallmouth bass exhibited lower $\dot{M}_{O_2TFM\,q0.8}$ (−0.15 to −0.26 μmol $O_2$ $g^{-1}$ $h^{-1}$ per gram of increase) and $\dot{M}_{O_2TFM\,Peak}$ (−0.44 to −1.01 μmol $O_2$ $g^{-1}$ $h^{-1}$ per gram of increase), but that was not the case for $\Delta\dot{M}_{O_2TFM\,Peak}$. Finally, smallmouth bass with a higher condition factor also exhibited lower $\dot{M}_{O_2TFM\,q0.8}$ (−0.15 to −0.29 μmol $O_2$ $g^{-1}$ $h^{-1}$ per 0.1 increase in condition) and $\dot{M}_{O_2TFM\,Peak}$ (−0.37 to −0.75 μmol $O_2$ $g^{-1}$ $h^{-1}$ per gram of increase), but that was not the case for either $\Delta\dot{M}_{O_2TFM\,q0.8}$ or $\Delta\dot{M}_{O_2TFM\,Peak}$.

### The relationship between MMR$_F$ and TFM-induced changes in $\dot{M}_{O_2}$ is species specific

In lake sturgeon, both $\dot{M}_{O_2TFM\,q0.8}$ and $\dot{M}_{O_2TFM\,Peak}$ during exposure to 8 mg $l^{-1}$ of TFM were significantly higher than MMR$_F$ (by 21% and 35%, respectively; GLM, F=19.0, *P*<0.001, followed by a Tukey test, *P*<0.001 for both comparisons; Fig. 4), but $\dot{M}_{O_2TFM\,q0.8}$ and $\dot{M}_{O_2TFM\,Peak}$ were not significantly different from each other (Tukey test, *P*=0.22). For rainbow trout, this relationship was inverted, with both $\dot{M}_{O_2TFM\,q0.8}$ and $\dot{M}_{O_2TFM\,Peak}$ during exposure to 60 mg $l^{-1}$ of TFM being significantly lower than MMR$_F$ (by 33% and 40%, respectively; GLM, F=95, *P*<0.001, followed by a Tukey test, *P*<0.001 for both comparisons; Fig. 4). Finally, the $\dot{M}_{O_2TFM\,q0.8}$ and $\dot{M}_{O_2TFM\,Peak}$ of smallmouth bass exposed to 60 mg $l^{-1}$ of TFM were not significantly different from their respective MMR$_F$ (GLM, F=1.14, *P*=0.34; Fig. 4). In all cases, the delta variables showed the same trends as the absolute variables.

### Lake sturgeon whose $\dot{M}_{O_2TFM}$ goes severely beyond the respective MMR$_F$ are more likely to perish during TFM exposure

The difference between MMR$_F$ and TFM-induced changes in oxygen consumption can be used to predict the likelihood

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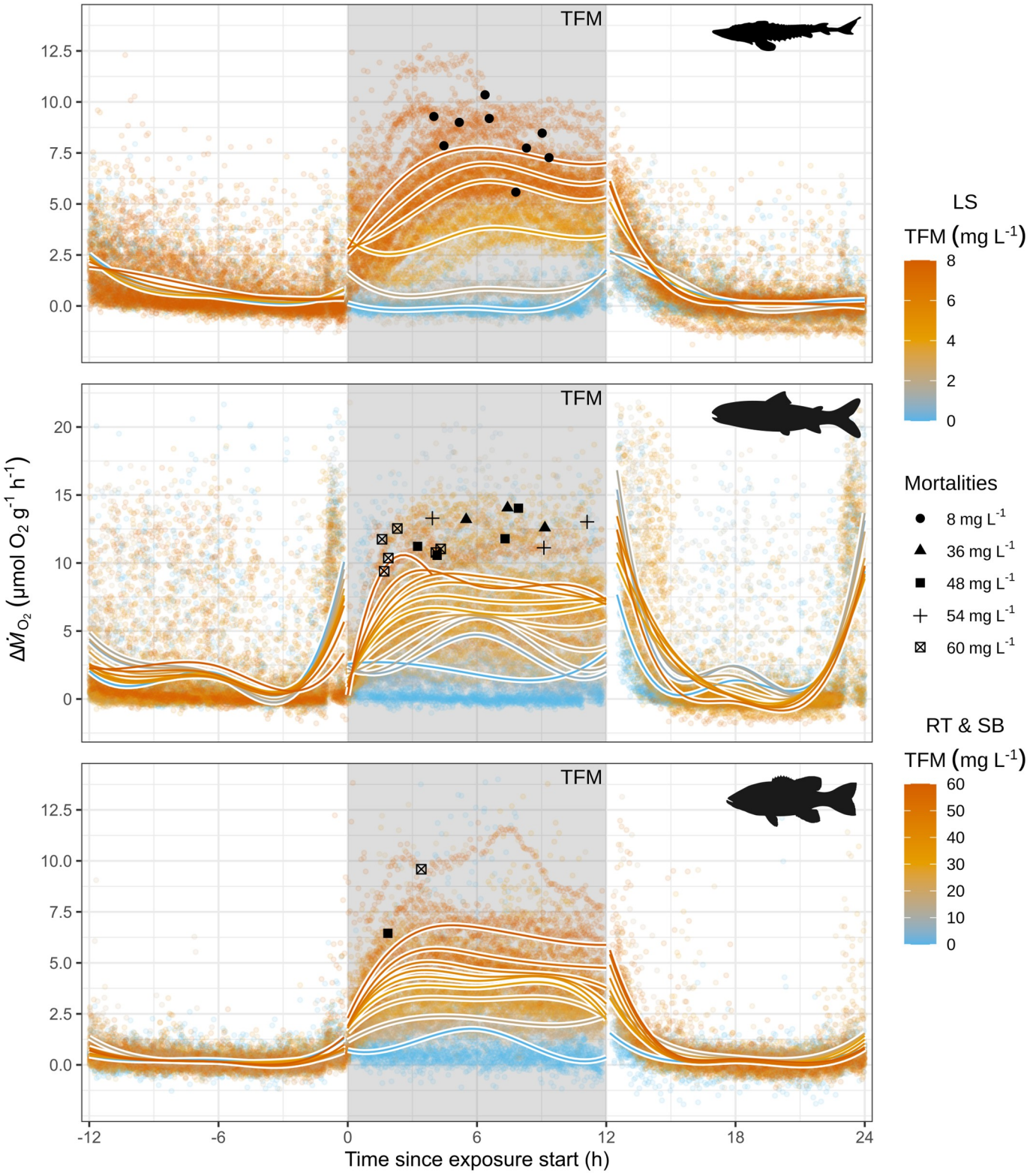


**Fig. 3. Delta oxygen consumption (absolute $\dot{M}_{O_2}$ minus the animal's respective SMR) of lake sturgeon (*n*=66), rainbow trout (*n*=70), and smallmouth bass (*n*=60) throughout the 25 TFM toxicity trials.** The colored lines show the generalized additive-modelled population trends over time for the various TFM concentrations tested. TFM exposures were conducted approximately at the same time of day every day (08:00-20:00) to minimize the effects of circadian rhythms. Eighteen $\Delta\dot{M}_{O_2}$ values for rainbow trout and smallmouth bass were omitted to optimize vertical plot space. Note the different Y-axis ranges per panel, and the different ranges in TFM concentration for lake sturgeon (LS, top color ramp) and rainbow trout and smallmouth bass (RT and SB, bottom color ramp).

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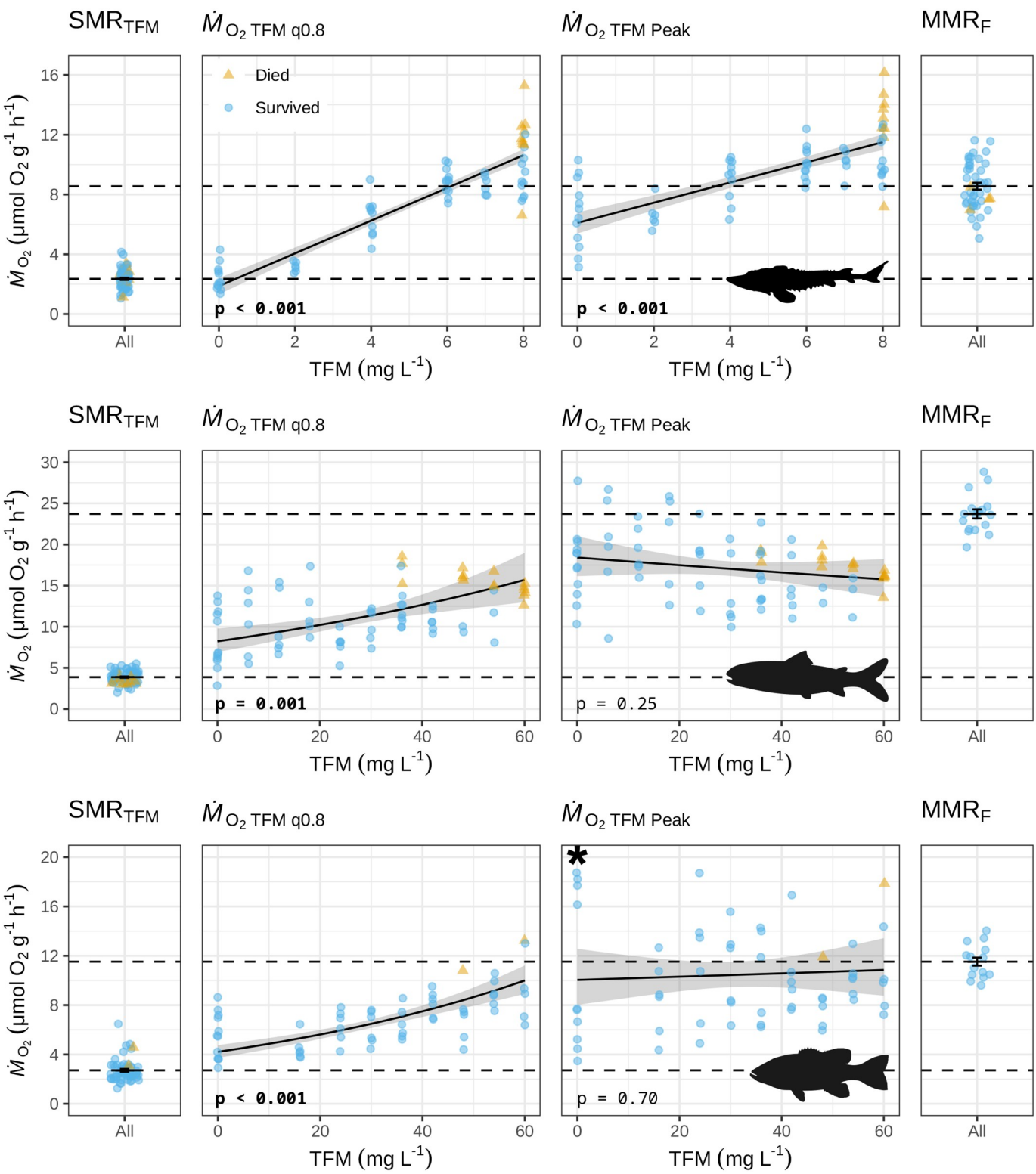


**Fig. 4. Comparison of the TFM-induced changes in oxygen consumption ($\dot{M}_{O_2\,TFM\,q0.8}$ and $\dot{M}_{O_2\,TFM\,Peak}$) to the SMR calculated prior to exposure ($SMR_{TFM}$) and to the MMR derived from chasing trials ($MMR_F$) for lake sturgeon, rainbow trout, and smallmouth bass.** The dashed black lines illustrate how the average $SMR_{TFM}$ and $MMR_F$ relate to the metrics calculated during TFM exposure. The solid black lines and respective ribbons show the fitted model and respective confidence intervals of the mean. *P*-values indicate the significance of the relationship between each metric and TFM concentration. For more details on the models applied, please refer to Tables 1, 2, and 3. *Two high values were omitted to optimize vertical plot space.

of death during exposure, with all the calculated differences being revealed as good predictors of the animals' likelihood of perishing during TFM exposure (GLM, *P*-values presented in Fig. 5 for simplicity). However, these results should be taken with caution given the small number of mortalities in the dataset.

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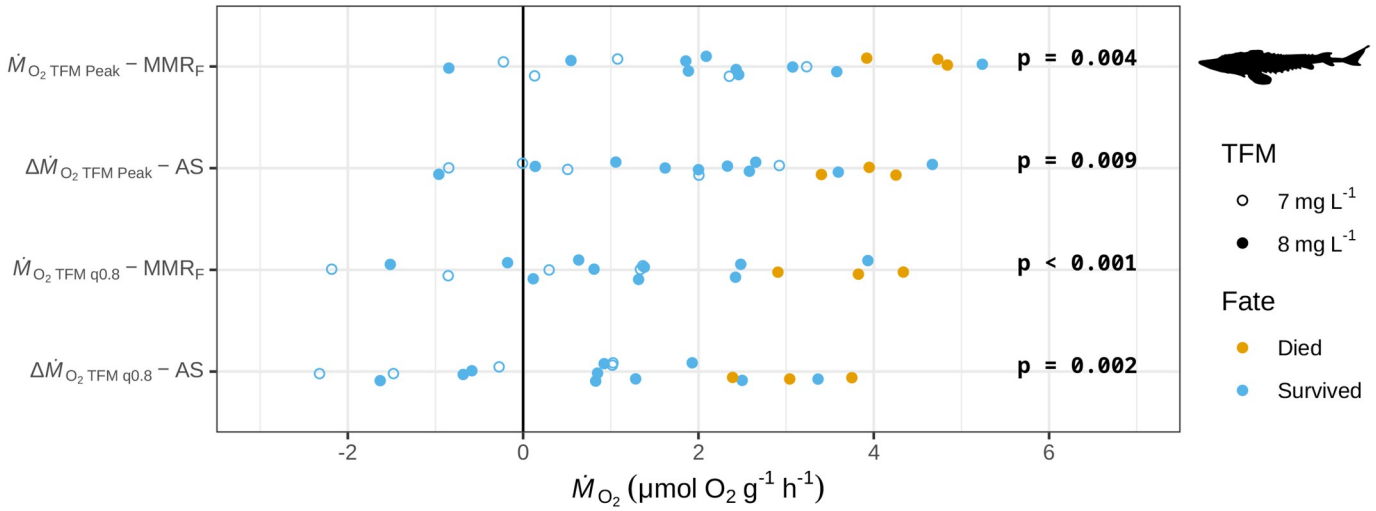


**Fig. 5. Difference between TFM-induced changes in oxygen consumption ($\dot{M}_{O_2\,TFM\,q0.8}$, $\Delta\dot{M}_{O_2\,TFM\,q0.8}$, $\dot{M}_{O_2\,TFM\,Peak}$, and $\Delta\dot{M}_{O_2\,TFM\,Peak}$) and respective exhaustive chasing MMRs ($MMR_F$ and AS) for lake sturgeon ($n$=18).** The black vertical line at $\dot{M}_{O_2} = 0$ denotes the point of no difference between the metrics calculated during TFM exposure and $MMR_F$/AS. *P*-values indicate the capacity of each of the metrics listed on the Y-axis to predict the lake sturgeon's fate during TFM exposure.

### Animals that survive TFM exposure quickly return to baseline oxygen consumption levels

In lake sturgeon, average $EPOC_{TFM}$ values ranged between 6.0 and 9.4 μmol $O_2$ $g^{-1}$ (for 2 and 8 mg $l^{-1}$ of TFM, respectively, Fig. 6, top left), with TFM concentration significantly increasing $EPOC_{TFM}$ (Table 2). Generally, $EPOC_{TFM}$ values for lake sturgeon were within the same range as $EPOC_{Chase}$, which averaged 7.6±0.6 μmol $O_2$ $g^{-1}$ ($n$=44). In rainbow trout, average $EPOC_{TFM}$ values ranged between 10.6 and 21.52 μmol $O_2$ $g^{-1}$ (for 54 and 24 mg $l^{-1}$ of TFM, respectively, Fig. 6, middle left). $EPOC_{TFM}$ displayed a non-linear trend but was not significantly related to the TFM concentration that the rainbow trout had been exposed to (Table 3) and was higher than their average $EPOC_{Chase}$ of 8.9±0.6 μmol $O_2$ $g^{-1}$ ($n$=18). Finally, in smallmouth bass, average $EPOC_{TFM}$ values ranged between 3.5 and 8.9 μmol $O_2$ $g^{-1}$ (for 42 and 54 mg $l^{-1}$ of TFM, respectively, Fig. 6, bottom left). Similarly to rainbow trout, the $EPOC_{TFM}$ of smallmouth bass was not significantly related to the TFM concentration that they had been exposed to (Table 4), but contrary to rainbow trout, $EPOC_{TFM}$ values for smallmouth bass were lower than their average $EPOC_{Chase}$ of 9.8±0.3 μmol $O_2$ $g^{-1}$ ($n$=18).

In lake sturgeon, average $t_{1/2\ TFM}$ values ranged between 2.1 and 5.7 h (for 6 and 2 mg $l^{-1}$ of TFM, respectively, Fig. 6, top right). $t_{1/2\ TFM}$ values for lake sturgeon were generally lower than the $t_{1/2\ Chase}$, which averaged 5.0±0.5 h ($n$=43). TFM concentration appeared to have a weak effect on $t_{1/2\ TFM}$, but upon closer inspection, that was deemed to be a pivoting effect of generally high $t_{1/2\ TFM}$ values for the animals exposed to 2 mg $l^{-1}$ of TFM. When those animals were removed from the model, the relationship clearly disappeared from the model (Table 2). In rainbow trout, average $t_{1/2\ TFM}$ values ranged between 1.2 and 4.5 h (for 12 and 42 mg $l^{-1}$ of TFM, respectively, Fig. 6, middle right). Similarly to lake sturgeon, $t_{1/2\ TFM}$ values for rainbow trout were generally lower than the $t_{1/2\ Chase}$, which averaged 3.7±0.3 h ($n$=18). There was a slight non-linear trend between TFM concentration and $t_{1/2\ TFM}$ for rainbow trout, with $t_{1/2\ TFM}$ increasing as TFM increased from 20 to 42 mg $l^{-1}$ but lowering at the highest concentrations (Table 3).

In smallmouth bass, average $t_{1/2\ TFM}$ values ranged between 2.0 and 5.7 h (for 36 and 16 mg $l^{-1}$ of TFM, respectively, Fig. 6, bottom right). As above, $t_{1/2\ TFM}$ values for smallmouth bass were generally lower than the $t_{1/2\ Chase}$, which averaged 4.3±0.5 h ($n$=14). Like with lake sturgeon, TFM concentration appeared to have a weak effect on $t_{1/2\ TFM}$, but that was deemed to be a pivoting effect of generally high $t_{1/2\ TFM}$ values for the animals exposed to 18 mg $l^{-1}$ of TFM (the lowest TFM concentration used for smallmouth bass). Upon removal of these pivotal data points from the model, the relationship between $t_{1/2}$ and TFM clearly disappeared (Table 4).

## DISCUSSION

With this study, we aimed to determine if non-target species exhibit stepwise increases in oxygen consumption ($\dot{M}_{O_2}$) during exposure to increasing concentrations of TFM and to determine if exercise-derived MMR was comparable to the oxygen consumption levels displayed during lethal TFM exposure in three non-target fish species with varying TFM sensitivities: lake sturgeon, rainbow trout, and smallmouth bass. Using intermittent-flow respirometry, we confirmed the presence of a stepwise relationship between TFM and oxygen consumption in all three non-target species. However, contrary to originally expected, the relationship between MMR and oxygen consumption during lethal TFM exposure was complex and species dependent. While this complex relationship broadens the need for species-specific testing, it reveals a link between the two variables that is useful for future testing of TFM sensitivity.

### TFM exposure leads to stepwise increases in $\dot{M}_{O_2}$ in non-target fishes

All three non-target fish species exhibited stepwise increases in $\dot{M}_{O_2}$ as a function of exposure to increasing TFM concentrations. As a protonophore, TFM increases the permeability of the mitochondria's inner membrane, disrupting the proton gradient that powers ATP synthesis. The weakening of the proton gradient leads to an increase in the activity of the ETC, which requires oxygen molecules as final receptors of electrons. Ultimately, TFM increases the amount of oxygen required to convert ADP to ATP,

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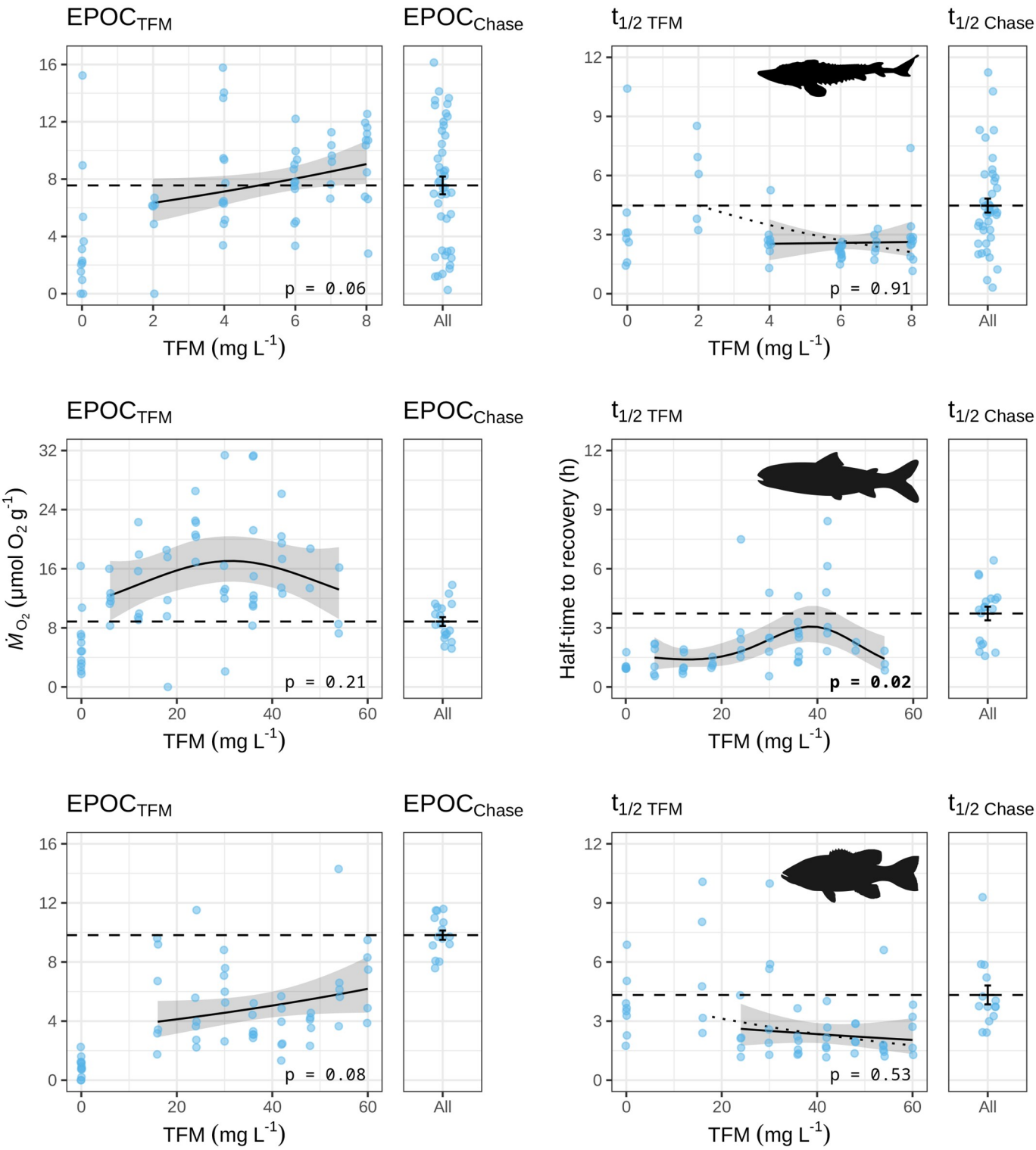


**Fig. 6. Excess oxygen consumption following TFM exposure and manual chasing (left) and half-time for oxygen consumption recovery (right) for lake sturgeon ($n_{Chase}$=44, $n_{TFM}$=59), rainbow trout ($n_{Chase}$=18, $n_{TFM}$=61), and smallmouth bass ($n_{Chase}$=14, $n_{TFM}$=58).** The dashed black lines illustrate how the chasing metrics compare to the TFM metrics. The solid black lines and respective ribbons show the fitted model and respective confidence intervals of the mean. *P*-values indicate the significance of the relationship between each metric and TFM concentration. For more details on the models applied, please refer to Tables 1-3. Dotted lines shown in the $t_{1/2}$ TFM panels for lake sturgeon and smallmouth bass indicate model fits if the lowest TFM concentration data were included in the model.

causing a metabolic imbalance (Birceanu et al., 2009). In the face of this insult, animals may respond by (1) biotransforming and neutralizing TFM (thus restoring the original oxygen-to-ATP cost; Kane et al., 1994), (2) switching to anaerobic energy production pathways (therefore reducing reliance on oxygen-powered ATP production; Wilkie et al., 2007), or (3) increasing oxygen uptake

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(i.e. paying the added oxygen cost; D'Souza et al., 2025). As such, the overall TFM tolerance of a species/life stage is determined by the combination of several axes of resistance, with differences in the activity and efficiency of the detoxification enzyme UDP-glucuronyltransferase (UDPGT) playing the main role (Kane et al., 1994). More tolerant species, like bluegill (a Centrarchid species closely related to smallmouth bass) and rainbow trout, have considerably higher UDPGT efficiency than sea lamprey. In comparison to sea lamprey, bluegill also exhibit significantly higher expression of detoxification genes and more responsive transcripts overall after exposure to TFM (Lawrence et al., 2022). The production and storage of these TFM-detoxifying enzymes is crucial for limiting internal concentrations of TFM and minimizing its effects on the mitochondria, likely allowing smallmouth bass and rainbow trout to display an exceptional resistance to TFM compared to sensitive species like sea lamprey and lake sturgeon.

Lake sturgeon were the most TFM-sensitive of the three tested species, with sensitivity levels comparable to those of larval sea lamprey (predicted LC99.9 of 6.5-7.5 mg $l^{-1}$ of TFM for the water chemistry tested here; Bills et al., 2003). The high sensitivity of young lake sturgeon to TFM is well documented (Applegate and King, 1962; Boogaard et al., 2003), and their $\dot{M}_{O_2}$ response to TFM exposure closely resembles that of the larval sea lamprey (D'Souza et al., 2025). Young-of-the-year lake sturgeon are in a particularly challenging position during TFM exposure because (1) they do not have as powerful a detoxification enzyme repertoire as other non-target species (Le Clair, 2014, as cited by Birceanu et al., 2021), (2) they show limited capacity for increasing oxygen consumption beyond basal levels (data presented here), and (3) they have not yet had the time to accumulate the energy stores required to fuel long bouts of anaerobic energy production (Ionescu et al., 2021). As such, it was expected that the lake sturgeon would show the highest TFM sensitivity levels of the three species tested here. As previously discussed, the UDPGT enzymes of highly tolerant species are more efficient than those of the lake sturgeon, and other tolerant species like bluegill exhibit significantly higher expression of detoxification genes and more responsive transcripts overall than larval sea lamprey (Kane et al., 1994; Lawrence et al., 2022). More active species, like rainbow trout and smallmouth bass, are also capable of sustaining high rates of oxygen consumption for longer periods of time than lower-activity species like the sedentary lake sturgeon. Lower availability and efficiency of TFM-detoxifying enzymes in combination with a poor ability to sustain high rates of oxygen consumption likely explain the high sensitivity observed in young-of-the-year lake sturgeon.

### $\dot{M}_{O_2\,TFM\,q0.8}$ is a robust metric for non-lethal TFM sensitivity tests

The 80th percentile of the oxygen consumption readings during TFM exposure ($\dot{M}_{O_2\,TFM\,q0.8}$ and $\Delta\dot{M}_{O_2\,TFM\,q0.8}$) reliably increased in a dose-dependent manner with increasing concentrations of TFM for all three species. $\dot{M}_{O_2\,TFM\,q0.8}$ is resistant to sporadic cycle-to-cycle variations in $\dot{M}_{O_2}$, proving to be an effective metric for visualizing TFM sensitivity. Peak oxygen consumption during TFM exposure ($\dot{M}_{O_2\,TFM\,Peak}$) appears to be too heavily influenced by cycle-to-cycle variations to be a reliable metric for measuring TFM sensitivity in highly active species. Similar to the findings of D'Souza et al. (2025) for larval sea lamprey, $\dot{M}_{O_2\,TFM\,Peak}$ (and $\Delta\dot{M}_{O_2\,TFM\,Peak}$) increased in a dose-dependent manner for lake sturgeon, but that was not the case for rainbow trout and smallmouth bass. This discrepancy may also be linked to the differing lifestyles of these species: lake sturgeon and larval sea lamprey are both low-activity, sedentary species, while rainbow trout and smallmouth bass are high-activity, predatory species. An active lifestyle may increase the likelihood of the rainbow trout and smallmouth bass exhibiting naturally high bouts of activity during trials, which would then drive $\dot{M}_{O_2\,TFM\,Peak}$ up regardless of the presence or absence of TFM. Ultimately, our results indicate that $\dot{M}_{O_2\,TFM\,q0.8}$ (and $\Delta\dot{M}_{O_2\,TFM\,q0.8}$) is a robust metric that works across a range of species-specific activity and TFM sensitivity levels, while $\dot{M}_{O_2\,TFM\,Peak}$ (and $\Delta\dot{M}_{O_2\,TFM\,Peak}$) is a good predictor of TFM sensitivity only in relatively sedentary species.

### The relationship between MMR and $\dot{M}_{O_2\,TFM}$ is species specific

Lake sturgeon exhibited higher oxygen consumption values during TFM exposure than after chasing. Together with the observed poor CE for lake sturgeon, this strongly indicates that manual chasing is not a suitable method of eliciting MMR in this species. Svendsen et al. (2014) found that exhaustive-chasing-derived MMR ($MMR_F$) and spontaneous MMR ($MMR_S$) were positively correlated in juvenile lake sturgeon, suggesting that their behavior and physiological capacity influenced both measurements of MMR. The suitability of different methods to elicit MMR in fishes has been discussed (e.g. Killen et al., 2017; Norin and Clark, 2016). A comprehensive review by Norin and Clark (2016) found that swim-flume respirometry is a reliable method for measuring MMR in strong sustained swimmers, while exhaustive chasing is more suitable for measuring the MMR of species that do not maintain swimming for prolonged periods of time. In contrast, a review by Killen et al. (2017) found overall differences in estimated MMR between methods of forced swimming, swim flume, and exhaustive chasing to be non-critical. Lake sturgeon are considered poor swimmers across burst- and sustained-swimming categories compared to other fishes (Peake et al., 1997), likely reflecting their reliance on other physiological and morphological traits that have helped them persist through millennia (Kieffer et al., 2001). Therefore, lake sturgeon exist in a gray area between methods of estimating MMR. In this case, choosing any method and performing it well may be the most suitable solution to this dilemma (Killen et al., 2017).

Contrary to lake sturgeon, juvenile rainbow trout exhibited lower oxygen consumption values during TFM exposure than after chasing. This was a surprising result, especially given that manual chasing may not elicit true MMR in salmonids (Raby et al., 2020). Importantly, should our chasing method have underestimated MMR, then the real difference between MMR and the oxygen consumption during lethal TFM exposure is even higher than reported here. We postulate that this difference is most likely driven by the differential oxygen consumption capacity of different tissues in rainbow trout. When TFM diffuses through the body, it impacts all of the animal's tissues. However, tissue-specific sensitivity to TFM can differ, with previous research having found that depletion of anaerobic energy stores in the brain, liver, and kidneys was more significant than in the muscle (Birceanu et al., 2009, 2014; Clifford et al., 2012; Wilkie et al., 2007). Therefore, critical failure of these essential systems would lead to death without inducing MMR in rainbow trout (because MMR is driven by exhaustion of the muscle tissues). Future studies could address variation in tissue-specific oxygen consumption under TFM exposure to help elucidate the underlying mechanisms leading to death in rainbow trout before they reach MMR.

Smallmouth bass showed a strong resistance to TFM, as expected from earlier work (Boogaard et al., 2003). Unfortunately, this high resistance, compounded by a limited availability of specimens,

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prevented us from confidently determining lethal TFM concentrations for this species within our experimental conditions. Nevertheless, at the highest TFM concentrations tested (where limited mortality started to occur), oxygen consumption during TFM exposure approached MMR, which would be similar to what D'Souza et al. (2025) suggested to be the case for larval sea lamprey. However, fully characterizing this relationship would require further research at even higher TFM concentrations. While this would be interesting to fully understand the mechanisms that govern the relationship between whole-animal oxygen consumption and TFM exposure in resistant species, it would likely be of little application in field applications of TFM, which are usually ∼80% lower than the higher concentrations tested here.

The individual-level difference between MMR and oxygen consumption during TFM exposure for lake sturgeon revealed a strong potential for predicting if the animal will survive the exposure period. Interestingly, because we have found chasing to be a poor estimate of true MMR in young-of-the-year lake sturgeon, it is possible that we could use the naturally occurring peaks in oxygen consumption as a baseline for the calculation of the difference induced by TFM exposure. This would eliminate the need for long-term tracking of individual animals in holding and also remove any confounding variables related to the progression of time. However, caution should be taken in interpreting the significance of these results due to our limited sample size for repeated measurements of MMR and TFM-induced changes and the presence of only three mortalities within the individual-comparison dataset. Further research with a larger sample size would be required to further reinforce or reassess the significance of our results.

### Non-target fishes quickly return to normal oxygen consumption after TFM exposure

All three non-target species quickly returned to SMR after being returned to freshwater, regardless of the TFM concentration they had been exposed to, which is similar to the behavior described by D'Souza et al. (2025) for larval sea lamprey. It is interesting that $t_{1/2}$ following TFM exposure was lower than $t_{1/2}$ following exhaustive chasing in all three species, highlighting the transient nature of the TFM's impacts on the animal's physiology. However, it is important to note that other long-term effects of TFM exposure have been identified. For example, long-lasting decreases in olfactory functions following TFM exposure may hinder feeding in non-target species (Sakamoto et al., 2016). Further research on the potentially far-reaching effects of TFM on the physiology of non-target animals is necessary to develop a more comprehensive understanding of its impact on non-target species and ecosystems.

### A non-lethal method for estimating TFM sensitivity

A non-lethal method of deriving TFM sensitivity could complement lethal toxicity testing and promote the sustainable use of experimental animals. Additionally, only exposing study animals to sublethal TFM concentrations may allow for their reuse in other experiments, helping to maximize the amount of scientific data obtained from a smaller number of study animals. Furthermore, non-lethal testing can also be applied to the real-time monitoring of local fauna during TFM applications in the field. This would entail collecting a sample of aquatic animals from the dosing location before the TFM treatment and placing them in a field-respirometry set-up, whereby their oxygen consumption can be measured using intermittent-flow respirometry. TFM application could then be monitored in real time through the oxygen consumption of these animals. Like a canary in a coal mine, a percentage of the exercise-derived MMR (or spontaneous MMR) of the sentinel animals can serve as a threshold that allows for early warning that the TFM application may be surpassing safe limits for the biological community within that habitat.

The species-specific nature of the relationship between MMR and oxygen consumption during lethal TFM exposure limits the broad application of this method across species. Therefore, improving the robustness of this method would require testing a range of Great Lakes species that differ in physiology, life history, and ecology. It would be particularly interesting to apply this method to native Great Lakes lamprey species, which have faced large declines in abundance as a result of multiple environmental and anthropogenic stressors, including TFM applications (Marsden and Siefkes, 2019; Neave et al., 2021; Schuldt and Goold, 1980).

Using a non-lethal testing method to monitor changes in a species' TFM sensitivity will be useful in maintaining TFM treatments safe in the face of global warming. In warmer temperatures, larval sea lamprey tolerance to TFM increases as their capacity for TFM elimination is enhanced through increased TFM metabolism and excretion (Hlina et al., 2021; Mitrovic et al., 2026; Muhametsafina et al., 2019). Therefore, current sea lamprey control measures may be undermined by warmer water temperatures that last further into the TFM dosing period (September-October; Schueller et al., 2024) or increased transient elevations in temperature during TFM applications, which ultimately alter the safe TFM exposure for non-target species (Lennox et al., 2020). This is particularly relevant as sea lamprey have a very wide thermal width, with AS steadily increasing up until 28°C (H. Flávio and M.P. Wilkie, unpublished, as shown in Wilkie et al., 2022), a factor that likely contributed to their successful invasion of the Great Lakes (Wilkie, 2025). Because the thermal tolerance of native species is likely lower than that of sea lamprey (e.g. the thermal optimum of young-of-the-year lake sturgeon is estimated to be ∼23°C, Koledin, 2024; but their critical thermal maximum may reach up to 34°C, Bugg et al., 2020), the safety margins between the dose required to kill larval sea lamprey and the dose tolerated by native species are likely to shrink in warmer conditions, and real-time monitoring will be essential to avoid non-target mortality events.

## Conclusion

Our study has confirmed that the dose-dependent relationship between whole-animal oxygen consumption and TFM exposure is not just limited to larval sea lamprey (the target species) but is also present in non-target fish species across a wide range of TFM sensitivity levels. Furthermore, we have unveiled species-dependent relationships between MMR and oxygen consumption during lethal TFM exposure that open interesting avenues for further research. Taken together, our results expand upon the preliminary work of D'Souza et al. (2025) and demonstrate that intermittent-flow respirometry is a viable non-lethal method for assessing TFM sensitivity in non-target species. Considering that the toxicity of TFM is highly dependent on water chemistry and temperature, future research should focus on developing a field-ready, real-time monitoring system that relies on the oxygen consumption response of sentinel organisms to determine the risk of overdose and prevent non-target mortality.

## MATERIALS AND METHODS

### Animal collection and holding

Young-of-the-year lake sturgeon ($n$=95, 7-13 cm, 1.6-7.3 g) were provided by the Ontario Aquaculture Research Centre, Guelph University.

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Hatchery-reared juvenile rainbow trout ($n$=95, 10-16 cm, 11-45 g) were kindly provided by Ocean Trout Canada Inc., located in Prince Edward Island. Wild juvenile smallmouth bass ($n$=78, 5-10 cm, 2-14 g) were captured by a team from Coastal Action via electrofishing in Nova Scotia. Animals were transferred to Dalhousie University by truck in aerated fish transport totes maintained at 15°C. Each species was housed in zebrafish racks equipped with 10 l tanks in the Aquatron Laboratory, Dalhousie University. Species were kept in separate tanks, and tank density was managed depending on the size of the animals (ten to 20 lake sturgeons, five to six rainbow trout, or three to six smallmouth bass per tank). Tanks were supplied with artificially prepared Great Lakes water (more details provided below) on a common recirculating aquaculture system, and tank temperature ranged from 14.5°C to 15.5°C. Smallmouth bass tanks were fitted with clear plastic dividers to avoid territorial aggression. Animals were fed once daily, where lake sturgeon and smallmouth bass were fed ad lib with a mixture of bloodworms and 0.5 or 1.0 mm salmonid pellets, respectively, and rainbow trout were fed a 1-3% body-weight ratio of 2.0 mm salmonid pellets. All experimental procedures were approved by the Dalhousie Animal Care Committee (Animal Use Protocol No. 25-045) and adhered to the guidelines of the Canadian Council of Animal Care.

### Water chemistry

Artificial Great Lakes water was prepared with a target alkalinity of 180 mg $l^{-1}$ of $CaCO_3$ and a pH of 8.5-8.6, representative of levels naturally found in rivers draining to the Great Lakes. Batches of ∼800 l of dechlorinated water from the city of Halifax were carbonated using compressed $CO_2$ for at least 24 h. $CaCO_3$ (BioShop, Canada), magnesium sulfate ($MgSO_4{\cdot}7H_2O$; BioShop, Canada), magnesium chloride ($MgCl_2{\cdot}6H_2O$; BioShop, Canada), and potassium chloride (KCl; BioShop, Canada) were then gradually mixed in. Once all the reagents had been mixed, the water was aerated for 24 h to remove excess $CO_2$ and return the water to the target pH. Alkalinity in the fish holding system was monitored regularly using a Hanna alkalinity meter (HI775) and ranged from 65 to 182 mg $l^{-1}$ of $CaCO_3$. pH in the holding system was monitored using a benchtop pH meter (AB315 pH/mV meter, equipped with a liquid-filled pH/ATC epoxy body combination electrode, Fisherbrand, Canada) and ranged between 8.4 and 8.7. Water temperature was monitored daily in the holding tanks using a KitchenAid temperature probe and ranged from 14.5°C to 15.5°C.

### Experimental set-up

This study used intermittent-flow respirometry, a common method of measuring oxygen consumption in aquatic animals (e.g. Hejlesen et al., 2024; Svendsen et al., 2016). Four different sizes of respirometers were used in this study. These were built using clear PVC piping and acrylic caps; the small model had an open chamber space of 4 cm inner diameter and 10 cm length (168.5 ml including tubing); the medium, large, and extra-large models had open spaces of 5.2 cm inner diameter and 10, 16, and 22 cm length, respectively (274.5, 403.6, and 532.9 ml including tubing, respectively). The small and medium models were used for lake sturgeon and smallmouth bass, while the large and extra-large models were used for rainbow trout. Respirometer volume/animal volume ratios ranged between 25 and 120 for lake sturgeon, between 20 and 88 for smallmouth bass, and between 11 and 35 for rainbow trout. All chambers were tested for leaks by filling them and placing them outside the water. Respirometers were covered with individual opaque sleeves during trials to minimize disturbance to the fish.

The respirometers were held in a 200 l water bath connected to two separate recirculation systems: one for clean fresh water and one for TFM water (Figs S1-S6). Two chillers (EcoPlus, 1/4 HP) maintained two aerated cold water sumps (100×45×15 cm; 75 l of water) at a temperature of 14.7°C. A custom-built temperature controller used water from these cold sumps to maintain a freshwater sump and a TFM water sump at a temperature of 15±0.1°C. Two four-channel aeration pumps (FEDOUR 256PH Aquarium Air Pump) were used to thoroughly aerate the water.

Each respirometer was equipped with an oxygen probe (OXFLOW-HS; PyroScience GmbH, Aachen, Germany). One out of every four respirometers was also equipped with a temperature probe (TDIP15; PyroScience GmbH, Aachen, Germany) and a pH probe (PHFLOW-PK8; PyroScience GmbH) to measure the temperature and pH of the chamber. All probes were connected to one of four PyroScience FireSting-PRO or $O_2$ optical meters (FSPRO-4; PyroScience GmbH, FSO2-C4; PyroScience GmbH, respectively), allowing in-chamber oxygen concentration and temperature to be measured every second and pH every 10 s. Bath temperature and pH were also recorded using a benchtop pH meter (AB315 pH/mV meter, equipped with a liquid-filled pH/ATC epoxy body combination electrode, Fisherbrand, Canada). These readings were relayed in real time to a custom-made R-based integration system, which monitored oxygen levels and temperature for unexpected changes. The integration system calculated oxygen consumption slopes and controlled the intermittent-flow phases based on $R^2$ thresholds and minimum acceptable oxygen levels (i.e. dynamic cycling). Each chamber was equipped with a recirculation pump (HISP-01; KOOKEE) and a flush pump (AD20P-0510A; SKESTORE). The flush pumps were controlled using a custom-built cycle controller (powered by Arduino microcontroller boards). With instruction from the cycle controller, the flush pumps alternated between phases of measurement (internal recirculation and measurement of oxygen levels) and flush (flushing of cooled, oxygenated water through the chamber). Measurement phases lasted a maximum of 240 s for rainbow trout and smallmouth bass and 480 s for lake sturgeon. The cycle controller terminates measurement phases early if 2 min has passed and the calculated $R^2$ exceeds 0.95 or if in-chamber oxygen concentration reaches 180 hPa (90-85% dissolved oxygen depending on atmospheric pressure conditions). Flush phases for all species lasted up to 120 s or until in-chamber oxygen levels stabilized. In each trial, either six animals of a single species or 12 animals of two different species were tested, always with one animal per respirometer chamber (i.e. six or 12 chambers were used simultaneously).

### Experimental procedures

This work reports the results of two types of experimental procedures (chasing trials and TFM exposure trials) performed using three different species (lake sturgeon, rainbow trout, and smallmouth bass). Experiments on rainbow trout and smallmouth bass were performed only at the population level (i.e. the individual identity of animals used in the two sets of trials was not tracked due to space constraints), while experiments for lake sturgeon were performed both at the population level and at the individual level, where a subset of the animals was identified across chasing and TFM exposure trials. Statistical tests were performed separately for the population-level analyses and for the individual-level analyses (see the data analysis section for further details). Where possible, animals of two species were chosen for each trial to maximize homogeneity in the experimental design.

#### Exhaustive chasing trials

Pre-experiment background bacterial $O_2$ consumption was measured for at least five cycles, each consisting of a 240-s measurement phase and a 60-s flush phase. Selected animals ($n$=12) were fasted 48 h prior to the experiment. After recording the background oxygen consumption, each animal was briefly weighed in a low volume of water on a precision scale to the nearest 0.01 g. Total length was measured to the nearest millimeter using a viewing box equipped with a ruler. The animal was then moved to the respirometer, and an opaque cover was placed on each respirometer to minimize visual disturbance. Measurement of the SMR for each animal took place for ∼24 h (more details on metric calculations provided in the data analysis section). After 24 h, MMR was measured by removing each fish from the respirometer one at a time and placing it into a 20 l bucket lined with bubble wrap. The smallmouth bass and rainbow trout were chased for 5 min while the lake sturgeon were chased for 7 min (determined as the lengths of time required to cause a loss of equilibrium in each species). Fish were swiftly returned to their respirometers (∼10 s post-exercise) to start the measurement process. Post-exercise $\dot{M}_{O_2}$ was then measured for 12 h.

After the 12-h recovery measurement period, animals were removed from the respirometers and placed into their holding tanks to recuperate for

later reuse in the TFM trials. At least five cycles of post-experiment background $O_2$ consumption were then measured before ending the trial. Trials were always initiated at approximately the same time of day to minimize potential impacts of circadian rhythms on the results.

### TFM exposure trials

Prior to trials, the oxygen and pH probes were calibrated, and background oxygen consumption levels were measured for at least five cycles. Each trial consisted of three 12-h periods: pre-exposure, exposure, and recovery. Similar to the chasing trials, six to 12 animals were fasted for approximately 48 h before each trial, then weighed in a film of water (nearest 0.01 g) and measured to the (nearest mm), before being placed in the respirometry chambers, which were covered to minimize disturbance. Oxygen consumption measurements began immediately following placement of the animals in their respective chambers, where they were left overnight for at least 12 h for measurements of pre-exposure SMR. To begin the exposure period, the water was dosed with a predetermined concentration of TFM. The animals were exposed to TFM for 12 h. Temperature, alkalinity, pH and TFM concentrations were closely monitored during exposure (Table S1). Oxygen consumption was constantly monitored during trials, and animals that appeared to be nearing death based on levels of oxygen consumption (i.e. displayed sharp declines in $\dot{M}_{O_2}$) were removed from their chambers and euthanized with an overdose of tricaine methanesulfonate (MS-222), which was buffered with sodium bicarbonate. At the end of the exposure period, the water bath was drained of TFM water, rinsed, and refilled with clean, artificial Great Lakes water. To avoid oxygen concentration within the rainbow trout respirometers from dropping to dangerous levels during the water-changing process, the rainbow trout respirometers were temporarily moved to a separate bath while the main system was rinsed and refilled, and the chambers were then returned to the main water bath once the system was ready. While this transfer was essential for the well-being of the animals, it did cause additional stress, which is visible as a high spike in metabolic activity at the beginning of the recovery period in Fig. 3. The animals were allowed to recover in clean water during a subsequent 12-h recovery period to quantify oxygen consumption levels during their recovery. After the recovery period, the animals were removed from the chambers and euthanized with an overdose of buffered MS-222. Post-trial background oxygen consumption was then measured for at least five cycles. This study was approved by the Animal Care and Use Program at Dalhousie University (AUP 25-045) and conducted according to the guidelines of the Canadian Council on Animal Care.

Small and medium respirometers were rinsed with clean water and sprayed with 70% ethanol after every trial, while large and extra-large respirometers were cleaned every one to two trials. At the end of every week, the whole experimental system was completely drained, scrubbed, sprayed with ethanol, and left to dry overnight.

### Individual comparison of chasing and TFM exposure

Exhaustive chasing and TFM exposure procedures were conducted for an individual-level comparison of 24 lake sturgeon, following the methods described in the 'Exhaustive chasing trials' and 'TFM exposure trials' sections. These lake sturgeon were tracked such that the same respirometer was used in both the chasing and TFM exposure trials to minimize potential effects of the equipment. The animals underwent a chasing trial and then were allowed to recover for 23 days before the start of the TFM exposure trials. This 23-day interval falls within the short-term 5-week range shown to produce repeatable estimates of metabolic rate metrics in juvenile brown trout (*Salmo trutta*; Norin and Malte, 2011) and allows the animals time to recover from the chasing trials before undergoing the TFM exposure trials.

## Calculations, statistics, and data analysis

### $\dot{M}_{O_2}$ calculations

$\dot{M}_{O_2}$ was calculated in R V4.5.2 (R Core Team, 2025, https://www.R-project.org/) using the R package pyroresp v0.1.2 (Flávio, 2026, https://github.com/hugomflavio/pyroresp). Recorded $O_2$ values (hPa) were converted to µmol $O_2$ $g^{-1}$ $h^{-1}$ using the R package respirometry v2.0.2 (Birk, 2025, https://doi.org/10.32614/CRAN.package.respirometry). Background respiration at the time of each experimental cycle was determined by linearly modeling the change in background respiration using pre- and post-background measurements of oxygen consumption. Estimated background oxygen slopes were subtracted from the calculated oxygen consumption slope of each cycle. For the chasing trials, pre-background measurements averaged 12.8±2.0%, 4.4±0.7%, and 12.2±2.3% of SMR, and post-background measurements averaged 13.6±1.3%, 11.8±2.4%, and 28.9±6.5% of SMR (mean±s.e.m. for lake sturgeon, rainbow trout, and smallmouth bass, respectively). For the TFM trials, pre-background measurements averaged 26.7±3.2, 3.0±0.5, and 11.7±2.8% of SMR, and post-background measurements averaged 20.1±1.7, 3.3±0.3, and 15.6±1.6% of SMR (mean±s.e.m. for lake sturgeon, rainbow trout, and smallmouth bass, respectively). To reduce the potential impacts of background on the variables of interest, animals with background levels above 100% of their respective SMR were excluded (amounting to four smallmouth bass and one lake sturgeon in the chasing trials and five lake sturgeons in the TFM trials). Cycles with an $R^2$ of 0.9 or higher were used in $\dot{M}_{O_2}$ determination. Oxygen slopes were converted into $\dot{M}_{O_2}$ using the equation $\dot{M}_{O_2} = S \times V \times M^{-1}$, where S is the corrected oxygen slope (µmol $O_2$ $g^{-1}$ $h^{-1}$), V is the volume of the respirometer corrected for the mass of the animal (assuming a 1 g:1 ml animal density), and M is the mass of the animal (g). These $\dot{M}_{O_2}$ values were then used to calculate a series of variables that are described in detail below and summarized in Table 1.

### Pre-chase/pre-exposure SMR calculations

SMR was calculated as the 20th percentile of the $\dot{M}_{O_2}$ measurements recorded prior to chasing ($SMR_{Chase}$) or prior to TFM exposure ($SMR_{TFM}$) using the *quantile()* function in R (Chabot et al., 2016). The pre-chasing and pre-exposure periods lasted approximately 24 and 12 h, over which over 400 and 200 $\dot{M}_{O_2}$ measurements were collected per individual, respectively (more details provided in the Results). The $SMR_{Chase}$ of one lake sturgeon was discarded, as the $R^2$ of most measurements failed to meet the 0.9 threshold. Similarly, the $SMR_{TFM}$ of one smallmouth bass was discarded as the chamber was found partly open during the trial.

Separate GLMs with Gamma distribution and log link function were applied to each species to determine if there were any effects of day since the first experiment, animal mass and condition, and background oxygen consumption on $SMR_{Chase}$ and $SMR_{TFM}$. Background was included as a variable in these and all subsequent models to account for potential biases introduced by higher background levels (up to 78%), which may artificially lower SMR estimates (H.F., unpublished observations). Model fitting was confirmed by inspecting Q-Q and residual plots using the R package gratia v0.11.2 (Simpson, 2024). Analysis of variance (Type III) (R package car; Fox and Weisberg, 2019) was applied to assess the significance of the tested variables. A Tukey post hoc test was performed when significant differences were found between variables (R package emmeans; Lenth and Piaskowski, 2026, https://doi.org/10.32614/CRAN.package.emmeans).

### MMR calculations

$MMR_F$ (induced by forced, exhaustive exercise) was determined as the post-chase cycle with the highest $\dot{M}_{O_2}$ measurement. Furthermore, AS was calculated by subtracting the $SMR_{Chase}$ of each individual from its respective $MMR_F$. A Gamma log-link GLM was applied to determine any effects of day since the first experiment, animal mass and condition, and background oxygen consumption on $MMR_F$. Model fitting and variable significance were tested as stated in 'Pre-chase/pre-exposure SMR calculations' section. Additionally, $MMR_S$ was determined as the cycle with the highest level of spontaneous activity (i.e. the highest $\dot{M}_{O_2}$ measurement) prior to chasing. This allowed assessing CE as a method to elicit MMR by calculating the difference between $MMR_S$ and $MMR_F$, as follows:

$$CE = (MMR_F - MMR_S)/MMR_S \times 100,$$

such that CE values equal or higher than 0 indicate that chasing elicits exhaustive oxygen consumption rates (i.e. $MMR_F$ equal or higher than $MMR_S$) and negative CE values indicate that chasing is a suboptimal

method to elicit exhaustive oxygen consumption rates (i.e. $MMR_F$ lower than $MMR_S$). One-sample *t*-tests were conducted to determine whether the CE value was significantly different from 0 for each species.

### Calculation of $\dot{M}_{O_2}$ changes induced by TFM exposure

Two different metrics were used to calculate the changes in $\dot{M}_{O_2}$ induced by TFM: 80th percentile of the exposure $\dot{M}_{O_2}$ readings ($\dot{M}_{O_2\,TFM\,q0.8}$) using the same *quantile()* function described for the calculation of SMR and the highest $\dot{M}_{O_2}$ value recorded during exposure ($\dot{M}_{O_2\,TFM\,Peak}$). Additionally, delta variants of these metrics ($\Delta\dot{M}_{O_2\,TFM\,q0.8}$ and $\Delta\dot{M}_{O_2\,TFM\,Peak}$) were calculated by subtracting the $SMR_{TFM}$ of each individual from each respective exposure variable. A full list of all variables calculated in this study is provided in Table 1. Decisions on the type of model to apply (GLM versus GAM) and distribution (Gaussian with identity link versus Gamma with log link) were made individually for each model depending on the nature of the response variable, the linearity of the relationship between the response variable and TFM, and the inspection of Q-Q and residual plots. In all GLMs, response variables encompass TFM concentration, time (number of days since the first trial for the species), background level (as percentage in relation to the animal's respective pre-exposure SMR), and animal mass and condition. In all GAMs, a smoother was applied to capture the non-linear effects of TFM concentration on $\dot{M}_{O_2}$, while time, background level, and animal mass and condition were included as fixed effects.

### Population-level comparison of MMR and TFM-induced changes in $\dot{M}_{O_2}$

$\dot{M}_{O_2\,TFM\,q0.8}$ and $\dot{M}_{O_2\,TFM\,Peak}$ at the highest TFM exposure were compared to $MMR_F$ with Gamma GLMs (log link), applied individually to each species, i.e. the model tested for differences in $\dot{M}_{O_2}$ values as a function of the variable type (factorial, three levels). Model fit and variable significance were confirmed as described above. Where significant differences were found, a Tukey post hoc test was used to determine the significance of the differences among paired groups. This analysis was repeated for the $\Delta\dot{M}_{O_2\,TFM\,q0.8}$, $\Delta\dot{M}_{O_2\,TFM\,Peak}$, and AS of each species.

### Individual-level comparison of $MMR_F$ and TFM-induced changes in $\dot{M}_{O_2}$ in lake sturgeon

Three lake sturgeons for which the difference in recorded mass between the chasing and TFM exposure trials was unrealistically negative (less than −0.9 g) were removed from the dataset, as these likely reflected an error in one of the weight measurements. Furthermore, two lake sturgeons were excluded because their chambers displayed very high background levels (>100% of SMR). As such, for 18 individually identified lake sturgeons, calculations of the absolute difference ($\dot{M}_{O_2\,TFM\,Peak} - MMR_F$; $\dot{M}_{O_2\,TFM\,q0.8} - MMR_F$) and delta difference ($\Delta\dot{M}_{O_2\,TFM\,Peak} - AS$; $\Delta\dot{M}_{O_2\,TFM\,q0.8} - AS$) were conducted. GLMs with Bernoulli distribution and logit link were then applied to test if the relative differences calculated above could be used as a predictor of survival to TFM exposure. TFM concentration was included as a predictive variable to account for the fact that some animals were exposed to 7 mg $l^{-1}$ of TFM (as opposed to 8 mg $l^{-1}$).

### Excess post-exercise/post-exposure oxygen consumption

Excess post-exercise oxygen consumption ($EPOC_{Chase}$) and excess post-exposure oxygen consumption ($EPOC_{TFM}$) were calculated as the area between post-chase $\dot{M}_{O_2}$ readings and the animal's respective SMR (Zhang et al., 2018), until $\dot{M}_{O_2}$ reached 1.1 times the SMR or 12 h had passed. Gamma GLMs (log link) were used to test if TFM concentration, time (number of days since the first trial for the species), background level (as percentage in relation to the animal's respective pre-exposure SMR), and animal mass and condition impacted $EPOC_{TFM}$. Control runs (i.e. TFM 0) were excluded from these models to improve their fit.

The half-time of oxygen consumption recovery following chasing ($t_{1/2\ Chase}$) or TFM exposure ($t_{1/2\ TFM}$) was calculated using the formulas provided by Endrenyi (1998). Summarily, the $\dot{M}_{O_2}$ values up until $\dot{M}_{O_2}$ reached 1.1 times the animal's respective SMR were log transformed and fitted with a linear regression. The slope of the linear regression (m) was then used to calculate k, which in turn allows the calculation of $t_{1/2}$, as follows:

$$k = -2.303 \times m,$$

$$t_{1/2} = \ln(2) \times k^{-1}.$$

Gamma GLMs (log link) were used to test if TFM concentration, time (number of days since the first trial for the species), background level (as percentage in relation to the animal's respective pre-exposure SMR), and animal mass and condition impacted $t_{1/2\ TFM}$. Control runs (i.e. TFM 0) were excluded from these models to improve model fit. Furthermore, for lake sturgeon and smallmouth bass, a second model was run, which excluded the lowest TFM concentration, to confirm the absence of a relationship between $t_{1/2\ TFM}$ and TFM concentration during exposure to higher TFM concentrations.

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#### Competing interests

The authors declare no competing or financial interests.

#### Author contributions

Conceptualization: R.L., H.F.; Formal analysis: M.L.G., A.M., H.F.; Funding acquisition: R.L., H.F.; Investigation: M.L.G., A.M., T.H., H.F.; Project administration: R.L., H.F.; Validation: M.L.G., A.M., H.F.; Visualization: M.L.G., A.M., H.F.; Writing – original draft: M.L.G., A.M., H.F.; Writing – review & editing: M.L.G., A.M., T.H., R.L., H.F.

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#### Data and resource availability

The respirometry data collected for this study and the respective R analysis scripts are available as a Zenodo repository: https://doi.org/10.5281/zenodo.22956090. All other relevant data and details of resources can be found within the article and its supplementary information.

#### First Person

This article has an associated First Person interview with the first author of the paper.

#### Peer review history

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#### References

**Applegate, V. C.** (1951). The sea lamprey in the great lakes. *The Scientific Monthly* **72**, 275-281. https://archive.org/details/sim_scientific-monthly_1951-05_72_5/page/n3/mode/2up

**Applegate, V. C. and King, E. L.Jr**. (1962). Comparative toxicity of 3-trifluormethyl-4-nitrophenol (TFM) to larval lampreys and eleven species of fishes. *Trans. Am. Fish. Soc.* **91**, 342-345. doi:10.1577/1548-8659(1962)91[342:CTOTTT]2.0.CO;2

**Bills, T. D., Boogaard, M. A., Johnson, D. A., Brege, D. C., Scholefield, R. J., Wayne Westman, R. and Stephens, B. E.** (2003). Development of a pH/alkalinity treatment model for applications of the lampricide TFM to streams tributary to the Great Lakes. *J. Great Lakes Res.* **29**, 510-520. doi:10.1016/S0380-1330(03)70512-7

**Birceanu, O., McClelland, G. B., Wang, Y. S. and Wilkie, M. P.** (2009). Failure of ATP supply to match ATP demand: the mechanism of toxicity of the lampricide, 3-trifluoromethyl-4-nitrophenol (TFM), used to control sea lamprey (*Petromyzon marinus*) populations in the Great Lakes. *Aquatic Toxicol.* **94**, 265-274. doi:10.1016/j.aquatox.2009.07.012

**Birceanu, O., McClelland, G. B., Wang, Y. S., Brown, J. C. L. and Wilkie, M. P.** (2011). The lampricide 3-trifluoromethyl-4-nitrophenol (TFM) uncouples mitochondrial oxidative phosphorylation in both sea lamprey (*Petromyzon marinus*) and TFM-tolerant rainbow trout (*Oncorhynchus mykiss*). *Comp. Biochem. Physiol. C Toxicol. Pharmacol.* **153**, 342-349. doi:10.1016/j.cbpc.2010.12.005

**Birceanu, O., Sorensen, L. A., Henry, M., McClelland, G. B., Wang, Y. S. and Wilkie, M. P.** (2014). The effects of the lampricide 3-trifluoromethyl-4-nitrophenol (TFM) on fuel stores and ion balance in a non-target fish, the rainbow trout (*Oncorhynchus mykiss*). *Comp. Biochem. Physiol. C Toxicol. Pharmacol.* **160**, 30-41. doi:10.1016/j.cbpc.2013.10.002

**Birceanu, O., Tessier, L. R., Huerta, B., Li, W., McDonald, A. and Wilkie, M. P.** (2021). At the intersection between toxicology and physiology: what we have learned about sea lampreys and bony fish physiology from studying the mode of action of lampricides. *J. Great Lakes Res.* **47**, S673-S689. doi:10.1016/j.jglr.2021.07.007

**Boogaard, M. A., Bills, T. D. and Johnson, D. A.** (2003). Acute toxicity of TFM and a TFM/niclosamide mixture to selected species of fish, including lake sturgeon (*Acipenser fulvescens*) and mudpuppies (*Necturus maculosus*), in laboratory and field exposures. *J. Gt. Lakes Res.* **29**, 529-541. doi:10.1016/S0380-1330(03)70514-0

**Borowiec, B. G., Birceanu, O., Wilson, J. M., McDonald, A. E. and Wilkie, M. P.** (2022). Niclosamide is a much more potent toxicant of mitochondrial respiration than TFM in the invasive sea lamprey (*Petromyzon marinus*). *Environ. Sci. Technol.* **56**, 4970-4979. doi:10.1021/acs.est.1c07117

**Bugg, W. S., Yoon, G. R., Schoen, A. N., Laluk, A., Brandt, C., Anderson, W. G. and Jeffries, K. M.** (2020). Effects of acclimation temperature on the thermal physiology in two geographically distinct populations of lake sturgeon (*Acipenser fulvescens*). *Conserv. Physiol.* **8**, coaa087. doi:10.1093/conphys/coaa087

**Bussy, U., Chung-Davidson, Y.-W., Buchinger, T., Li, K., Smith, S. A., Jones, A. D. and Li, W.** (2018). Metabolism of a sea lamprey pesticide by fish liver enzymes part A: identification and synthesis of TFM metabolites. *Anal. Bioanal. Chem.* **410**, 1749-1761. doi:10.1007/s00216-017-0830-8

**Chabot, D., Steffensen, J. F. and Farrell, A. P.** (2016). The determination of standard metabolic rate in fishes. *J. Fish Biol.* **88**, 81-121. doi:10.1111/jfb.12845

**Clifford, A. M., Henry, M., Bergstedt, R., McDonald, D. G., Smits, A. S. and Wilkie, M. P.** (2012). Recovery of larval sea lampreys from short-term exposure to the pesticide 3-trifluoromethyl-4-nitrophenol: implications for sea lamprey control in the great lakes. *Trans. Am. Fish. Soc.* **141**, 1697-1710. doi:10.1080/00028487.2012.713887

**D'Souza, L., Flávio, H. and Wilkie, M. P.** (2025). The lampricide 3-trifluoromethyl-4-nitrophenol (TFM) stimulates oxygen consumption by larval sea lamprey in a dose-dependent manner. *J. Gt. Lakes Res.* **51**, 102536. doi:10.1016/j.jglr.2025.102536

**Endrenyi, L.**, 1998. Pharmacokinetics. In *Principles of Medical Pharmacology* (ed. H. Kalant and W. H. E. Roschlau). Oxford University Press, New York, pp. 55-66.

**Eshenroder, R. L.** (2014). The role of the Champlain Canal and Erie Canal as putative corridors for colonization of Lake Champlain and Lake Ontario by sea lampreys. *Trans. Am. Fish. Soc.* **143**, 634-649. doi:10.1080/00028487.2013.879818

**Fox, J. and Weisberg, S.** (2019). *An R Companion to Applied Regression*, 3rd edn. Sage. https://www.john-fox.ca/Companion/.

**Gaden, M., Brant, C., Stedman, R. C., Cooke, S. J., Young, N., Lauber, T. B., Nguyen, V. M., Connelly, N. A. and Knuth, B.** (2021). Shifting baselines and social license to operate: Challenges in communicating sea lamprey control. *J. Great Lakes Re.* **47**, S800-S808. doi:10.1016/j.jglr.2021.01.016

**Hejlesen, R., Scheffler, F. B., Byrge, C. G., Kjær-Sørensen, K., Oxvig, C., Fago, A. and Malte, H.** (2024). Assessing metabolic rates in zebrafish using a 3D-printed intermittent-flow respirometer and swim tunnel system. *Biol. Open* **13**, bio060375. doi:10.1242/bio.060375

**Hlina, B. L., Birceanu, O., Robinson, C. S., Dhiyebi, H. and Wilkie, M. P.** (2021). The relationship between thermal physiology and lampricide sensitivity in larval sea lamprey (*Petromyzon marinus*). *J. Great Lakes Res.* **47**, S272-S284. doi:10.1016/j.jglr.2021.10.002

**Huerta, B., Chung-Davidson, Y.-W., Bussy, U., Zhang, Y., Bazil, J. N. and Li, W.** (2020). Sea lamprey cardiac mitochondrial bioenergetics after exposure to TFM and its metabolites. *Aquat. Toxicol.* **219**, 105380. doi:10.1016/j.aquatox.2019.105380

**Hunn, J. B. and Allen, J. L.** (1974). Movement of drugs across the gills of fishes. *Annu. Rev. Pharmacol. Toxicol.* **14**, 47-54. doi:10.1146/annurev.pa.14.040174.000403

**Ionescu, R. A., Hepditch, S. L. J. and Wilkie, M. P.** (2021). The lampricide 3-trifluoromethyl-4-nitrophenol causes temporary metabolic disturbances in juvenile lake sturgeon (*Acipenser fulvescens*): implications for sea lamprey control and fish conservation. *Conserv. Physiol.* **9**, coab069. doi:10.1093/conphys/coab069

**Kane, A. S., Kahng, M. W., Reimschuessel, R., Nhamburo, P. T. and Lipsky, M. M.** (1994). UDP-glucuronyltransferase kinetics for 3-trifluoromethyl-4-nitrophenol (TFM) in fish. *Trans. Am. Fish. Soc.* **123**, 217-222. doi:10.1577/1548-8659(1994)123<0217:UGKFTN>2.3.CO;2

**Kawatski, J. A., Dawson, V. K. and Reuvers, M. L.** (1974). Effect of TFM and Bayer 73 on in vivo oxygen consumption of the aquatic midge *Chironomus Tentans*. *Trans. Am. Fish. Soc.* **103**, 551-556. doi:10.1577/1548-8659(1974)103<551:EOTABO>2.0.CO;2

**Kieffer, J. D., Wakefield, A. M. and Litvak, M. K.** (2001). Juvenile sturgeon exhibit reduced physiological responses to exercise. *J. Exp. Biol.* **204**, 4281-4289. doi:10.1242/jeb.204.24.4281

**Killen, S. S., Norin, T. and Halsey, L. G.** (2017). Do method and species lifestyle affect measures of maximum metabolic rate in fishes? *J. Fish Biol.* **90**, 1037-1046. doi:10.1111/jfb.13195

**Koledin, M.** (2024). The effects of warmer temperatures on the physiology of invasive sea lamprey and imperiled populations of lake sturgeon in the Laurentian Great Lakes. *MSc thesis*, Wilfrid Laurier University. https://scholars.wlu.ca/etd/2634/.

**Lawrence, M. J., Grayson, P., Jeffrey, J. D., Docker, M. F., Garroway, C. J., Wilson, J. M., Manzon, R. G., Wilkie, M. P. and Jeffries, K. M.** (2022). Variation in the transcriptome response and detoxification gene diversity drives pesticide tolerance in fishes. *Environ. Sci. Technol.* **56**, 12137-12147. doi:10.1021/acs.est.2c00821

**Le Clair, M. W.** (2014). Distribution and elimination of 3-trifluoromethyl-4-nitrophenol (TFM) by sea lamprey (*Petromyzon marinus*) and non-target, rainbow trout (*Oncorhynchus mykiss*) and lake sturgeon (*Acipenser fulvescens*). *MSc thesis*, Wilfrid Laurier University.

**Lech, J. J. and Costrini, N. V.** (1972). In vitro and in vivo metabolism of 3-trifluoromethyl-4-nitrophenol (TFM) in rainbow trout. *Comp. General Pharmacol.* **3**, 160-166. doi:10.1016/0010-4035(72)90021-3

**Lech, J. J. and Statham, C. N.** (1975). Role of glucuronide formation in the selective toxicity of 3-trifluoromethyl-4-nitrophenol (TFM) for the sea lamprey: comparative aspects of TFM uptake and conjugation in sea lamprey and rainbow trout. *Toxicol. Appl. Pharmacol.* **31**, 150-158. doi:10.1016/0041-008X(75)90063-0

**Lennox, R. J., Bravener, G. A., Lin, H.-Y., Madenjian, C. P., Muir, A. M., Remucal, C. K., Robinson, K. F., Rous, A. M., Siefkes, M. J., Wilkie, M. P. et al.** (2020). Potential changes to the biology and challenges to the management of invasive sea lamprey *Petromyzon marinus* in the Laurentian Great Lakes due to climate change. *Glob. Change Biol.* **26**, 1118-1137. doi:10.1111/gcb.14957

**Marsden, J. E. and Siefkes, M. J.** (2019). Control of invasive sea lamprey in the Great Lakes, Lake Champlain, and Finger Lakes of New York. In *Lampreys: Biology, Conservation and Control: Volume 2* (ed. M. F. Docker), pp. 411-479. Springer Netherlands. doi:10.1007/978-94-024-1684-8_5

**McDonald, D. G. and Kolar, C. S.** (2007). Research to guide the use of lampricides for controlling sea lamprey. *J. Gt. Lakes Res.* **33**, 20-34. doi:10.3394/0380-1330(2007)33[20:RTGTUO]2.0.CO;2

**Mitrovic, D., Bragg, L. M., Flávio, H., Servos, M. R. and Wilkie, M. P.** (2026). The mechanism of increased tolerance to the pesticide 3-trifluoromethyl-4-nitrophenol (TFM) in invasive sea lamprey (*Petromyzon marinus*) at warmer water temperatures. *Environ. Toxicol. Chem.* **45**, 2193-2208. doi:10.1093/etojnl/vgag040

**Muhametsafina, A., Birceanu, O., Hlina, B. L., Tessier, L. R. and Wilkie, M. P.** (2019). Warmer waters increase the larval sea lamprey's (*Petromyzon marinus*) tolerance to the lampricide 3-trifluoromethyl-4-nitrophenol (TFM). *J. Gt. Lakes Res.* **45**, 921-933. doi:10.1016/j.jglr.2019.07.011

**Neave, F. B., Booth, R. M. W., Philipps, R. R., Keffer, D. A., Bravener, G. A. and Coombs, N.** (2021). Changes in native lamprey populations in the Great Lakes since the onset of sea lamprey (*Petromyzon marinus*) control. *J. Great Lakes Res.* **47**, S378-S387. doi:10.1016/j.jglr.2020.10.005

**Norin, T. and Clark, T. D.** (2016). Measurement and relevance of maximum metabolic rate in fishes. *J. Fish Biol.* **88**, 122-151. doi:10.1111/jfb.12796

**Norin, T. and Malte, H.** (2011). Repeatability of standard metabolic rate, active metabolic rate and aerobic scope in young brown trout during a period of moderate food availability. *J. Exp. Biol.* **214**, 1668-1675. doi:10.1242/jeb.054205

**Peake, S., Beamish, F. W. H., McKinley, R. S., Scruton, D. A. and Katopodis, C.** (1997). Relating swimming performance of lake sturgeon, Acipenser fulvescens, to fishway design. *Canadian Journal of Fisheries and Aquatic Sciences* **54**, 1361-1366. doi:10.1139/f97-039

**Pirie, E., Serville-Tertullien, M., Buell, M.-C., Wall, B. M. and Furgal, C.** (2025). Indigenous peoples and environmental research and monitoring within the Laurentian Great Lakes Basin: a systematic map. *Ecol. Solutions Evid.* **6**, e70102. doi:10.1002/2688-8319.70102

**Raby, G. D., Doherty, C. L. J., Mokdad, A., Pitcher, T. E. and Fisk, A. T.** (2020). Post-exercise respirometry underestimates maximum metabolic rate in juvenile salmon. *Conserv. Physiol.* **8**, coaa063. doi:10.1093/conphys/coaa063

**Sakamoto, K., Dew, W. A., Hecnar, S. J. and Pyle, G. G.** (2016). Effects of lampricide on olfaction and behavior in young-of-the-year lake sturgeon (*Acipenser fulvescens*). *Environ. Sci. Technol.* **50**, 3462-3468. doi:10.1021/acs.est.6b01051

**Schueller, J. R., Boogaard, M. A., Kirkeeng, C. A., Schloesser, N. A., Wolfe, S. L., Lettenberger, A. J., King-Heiden, T. C. and Luoma, J. A.** (2024).

Seasonal differences in larval sea lamprey (*Petromyzon marinus*) sensitivity to the pesticide TFM. *J. Gt. Lakes Res.* **50**, 102248. doi:10.1016/j.jglr.2023.102248

**Schuldt, R. J. and Goold, R.** (1980). Changes in the distribution of native lampreys in Lake Superior tributaries in response to sea lamprey (*Petromyzon marinus*) control, 1953–77. *Can. J. Fish. Aquat. Sci.* **37**, 1872-1885. doi:10.1139/f80-229

**Siefkes, M. J.** (2017). Use of physiological knowledge to control the invasive sea lamprey (*Petromyzon marinus*) in the Laurentian Great Lakes. *Conserv. Physiol.* **5**, cox031. doi:10.1093/conphys/cox031

**Simpson, G. L.** (2024). gratia: an R package for exploring generalized additive models. *J. Open Source Software* **9**, 6962. doi:10.21105/joss.06962

**Smith, B. R. and Tibbles, J. J.** (1980). Sea lamprey (*Petromyzon marinus*) in Lakes Huron, Michigan, and Superior: history of invasion and control, 1936–78. *Can. J. Fish. Aquat. Sci.* **37**, 1780-1801. doi:10.1139/f80-222

**Sullivan, W. P., Burkett, D. P., Boogaard, M. A., Criger, L. A., Freiburger, C. E., Hubert, T. D., Leistner, K. G., Morrison, B. J., Nowicki, S. M., Robertson, S. N. P. et al.** (2021). Advances in the use of lampricides to control sea lampreys in the Laurentian Great Lakes, 2000–2019. *J. Great Lakes Res.* **47**, S216-S237. doi:10.1016/j.jglr.2021.08.009

**Svendsen, J. C., Genz, J., Anderson, W. G., Stol, J. A., Watkinson, D. A. and Enders, E. C.** (2014). Evidence of circadian rhythm, oxygen regulation capacity, metabolic repeatability and positive correlations between forced and spontaneous maximal metabolic rates in lake sturgeon *Acipenser fulvescens*. *PLoS ONE* **9**, e94693. doi:10.1371/journal.pone.0094693

**Svendsen, M. B. S., Bushnell, P. G. and Steffensen, J. F.** (2016). Design and setup of intermittent-flow respirometry system for aquatic organisms: how to set up an aquatic respirometry system. *J. Fish Biol.* **88**, 26-50. doi:10.1111/jfb.12797

**Viant, M. R., Walton, J. H. and Tjeerdema, R. S.** (2001). Comparative sublethal actions of 3-trifluoromethyl-4-nitrophenol in marine molluscs as measured by *in vivo*$^{31}$P NMR. *Pestic. Biochem. Physiol.* **71**, 40-47. doi:10.1006/pest.2001.2554

**Wilkie, M. P.** (2025). How the resilient ecophysiology of the sea lamprey allowed them to invade the Laurentian Great Lakes and could protect them from climate change. *Front. Fish Sci.* **3**. doi:10.3389/frish.2025.1591515

**Wilkie, M. P., Holmes, J. A. and Youson, J. H.** (2007). The lampricide 3-trifluoromethyl-4-nitrophenol (TFM) interferes with intermediary metabolism and glucose homeostasis, but not with ion balance, in larval sea lamprey (*Petromyzon marinus*). *Can. J. Fish. Aquat. Sci.* **64**, 1174-1182. doi:10.1139/f07-082

**Wilkie, M. P., Hubert, T. D., Boogaard, M. A. and Birceanu, O.** (2019). Control of invasive sea lampreys using the piscicides TFM and niclosamide: toxicology, successes & future prospects. *Aquat. Toxicol.* **211**, 235-252. doi:10.1016/j.aquatox.2018.12.012

**Wilkie, M. P., Johnson, N. S. and Docker, M. F.** (2022). Invasive species control and management: the sea lamprey story. In *Fish Physiology*, Vol. 39 (ed. N. A. Fangue, S. J. Cooke, A. P. Farrell, C. J. Brauner and E. J. Eliason), pp. 489-579. Academic Press. doi:10.1016/bs.fp.2022.09.001

**Zhang, Y., Claireaux, G., Takle, H., Jørgensen, S. M. and Farrell, A. P.** (2018). A three-phase excess post-exercise oxygen consumption in Atlantic salmon *Salmo salar* and its response to exercise training. *J. Fish Biol.* **92**, 1385-1403. doi:10.1111/jfb.13593

**Table S1. Summary information of all the experimental trials conducted in this project.**

| Date | Exp. type | Species | Temperature | Alkalinity | pH | $TFM_{nominal}$ | $TFM_{observed}$ |
|---|---|---|---|---|---|---|---|
| 2025-11-09 | MMR | RT, SB | 14.8 (14.6-15.0) | n.a. | n.a. | - | - |
| 2025-11-11 | MMR | RT, SB | 15.1 (15.0-15.1) | 75 | n.a. | - | - |
| 2025-11-13 | MMR | RT, SB | 15.1 (15.0-15.2) | 81 | n.a. | - | - |
| 2025-11-16 | MMR | LS | 15.0 (14.9-15.2) | 92 | n.a. | - | - |
| 2025-11-18 | MMR | LS | 15.0 (14.9-15.0) | 120 | n.a. | - | - |
| 2025-11-20 | MMR | LS | 15.0 (14.9-15.0) | 177 | n.a. | - | - |
| 2025-11-23 | TFM | LS | 15.0 (14.9-15.1) | 175 | 8.58 (8.56-8.60) | 0 | 0 |
| 2025-11-25 | MMR | LS | 15.0 (14.9-15.0) | 177 | 8.56 (8.49-8.59) | - | - |
| 2025-11-27 | TFM | LS | 15.0 (14.7-15.1) | 180 | 8.54 (8.53-8.55) | 8 | 8.1 (7.9-8.3) |
| 2025-11-30 | TFM | LS | 15.0 (14.8-15.1) | 177 | 8.61 (8.58-8.62) | 4 | 4.1 (4.0-4.2) |

**Table S1. Continued.**

| Date | Exp. type | Species | Temperature | Alkalinity | pH | $TFM_{nominal}$ | $TFM_{observed}$ |
|---|---|---|---|---|---|---|---|
| 2025-12-02 | TFM | RT | 14.9 (14.7-15.0) | 186 | 8.53 (8.45-8.57) | 12 | 12.0 (11.9-12.3) |
| 2025-12-04 | TFM | RT | 14.9 (14.7-15.1) | 196 | 8.46 (8.41-8.50) | 18 | 18.1 (17.6-18.3) |
| 2025-12-07 | TFM | RT, SB | 14.9 (14.8-15.0) | 185 | 8.66 (8.63-8.67) | 0 | 0 |
| 2025-12-09 | TFM | LS, RT | 14.9 (14.5-15.0) | 180 | 8.56 (8.52-8.59) | 6 | 6.0 (5.8-6.3) |
| 2025-12-14 | TFM | RT, SB | 15.0 (14.9-15.0) | 184 | 8.63 (8.59-8.66) | 0 | 0 |
| 2025-12-16 | TFM | RT, SB | 15.0 (14.8-15.0) | n.a. | 8.55 (8.51-8.58) | 36 | 36.1 (35.6-36.6) |
| 2025-12-18 | TFM | RT, SB | 15.0 (14.9-15.1) | 185 | 8.54 (8.49-8.57) | 42 | 42.0 (40.7-43.0) |
| 2026-01-04 | TFM | RT, SB | 15.1 (14.9-15.3) | 182 | 8.57 (8.55-8.59) | 30 | 30.1 (28.9-32.1) |
| 2026-01-06 | TFM | RT, SB | 15.0 (14.8-15.2) | 182 | 8.54 (8.48-8.57) | 48 | 47.9 (46.9-48.6) |
| 2026-01-08 | TFM | RT, SB | 15.0 (14.9-15.1) | 176 | 8.54 (8.50-8.55) | 54 | 54.1 (53.2-55.9) |
| 2026-01-11 | TFM | RT, SB | 15.0 (14.9-15.1) | 177 | 8.37 (8.32-8.40)* | 24 | 24.2 (23.9-24.7) |
| 2026-01-13 | TFM | RT | 15.0 (14.9-15.0) | n.a. | 8.36 (8.31-8.40)* | 36 | 36.0 (34.9-37.0) |
| 2026-01-15 | TFM | RT, SB | 15.0 (14.9-15.1) | 185 | 8.40 (8.35-8.44)* | 60 | 59.3 (57.9-60.6) |

* The pH meter readings were lost for these exposure periods. The data from the sensor spots is available, but the high concentration of TFM in these trials caused considerable interference in the sensor spot readings during the exposure period. Therefore, the best approximation available, listed in these cells, is the summary of the sensor-spot readings in the pre-exposure period. These are likely slightly below the values that would have been recorded by the pH meter (by approximately 0.2 based on the remaining data) as the sensor spots gradually lost sensitivity with consecutive exposures to high concentrations of TFM.

**Table S1. Continued.**

| Date | Exp. type | Species | Temperature | Alkalinity | pH | $TFM_{nominal}$ | $TFM_{observed}$ |
|---|---|---|---|---|---|---|---|
| 2026-01-18 | TFM | LS | 15.0 (14.9-15.1) | 185 | 8.62 (8.58-8.64) | 6 | 5.9 (5.7-6.0) |
| 2026-01-20 | TFM | SB | 15.0 (14.9-15.2) | 191 | 8.55 (8.50-8.57) | 16 | 15.99 |
| 2026-01-25 | MMR | LS | 15.1 (14.6-15.2) | 177 | 8.55 (8.45-8.58) | - | - |
| 2026-01-27 | MMR | LS | 15.0 (15.0-15.1) | 177 | 8.53 (8.51-8.55) | - | - |
| 2026-01-29 | MMR | LS | 15.0 (14.9-15.1) | 182 | 8.52 (8.48-8.57) | - | - |
| 2026-02-01 | MMR | LS | 15.0 (14.9-15.0) | 176 | 8.64 (8.62-8.65) | - | - |
| 2026-02-03 | TFM | LS | 14.9 (14.8-15.0) | 184 | 8.54 (8.51-8.55) | 0 | 0 |
| 2026-02-05 | TFM | LS | 15.0 (14.7-15.1) | 186 | 8.51 (8.47-8.53) | 4 | 3.99 |
| 2026-02-15 | TFM | LS | 15.0 (14.9-15.1) | 180 | 8.61 (8.57-8.65) | 2 | 1.98 |
| 2026-02-17 | TFM | LS | 15.0 (14.9-15.1) | 181 | 8.42 (8.38-8.44) | 7 | 6.97 |
| 2026-02-19 | TFM | LS | 15.0 (14.9-15.1) | 179 | 8.57 (8.54-8.59) | 8 | 7.99 |
| 2026-02-22 | TFM | LS | 15.1 (15.0-15.3) | n.a. | 8.59 (8.54-8.62) | 8 | 8.09 |
| 2026-02-24 | TFM | LS | 15.1 (15.0-15.1) | 181 | 8.52 (8.49-8.53) | 8 | 8.05 |

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**Supplementary Material 2: Photos/schematics of the experimental system**

The experimental system design followed that described in Supplementary Material 1 of D'Souza et al (2025, doi: 10.1016/j.jglr.2025.102536). Summarily, the system consists of two independent recirculation systems, one running clean water and the other dosed with TFM, as shown in the schematic presented in the next page. Some photos are also provided.

**Schematic of the wet table:**

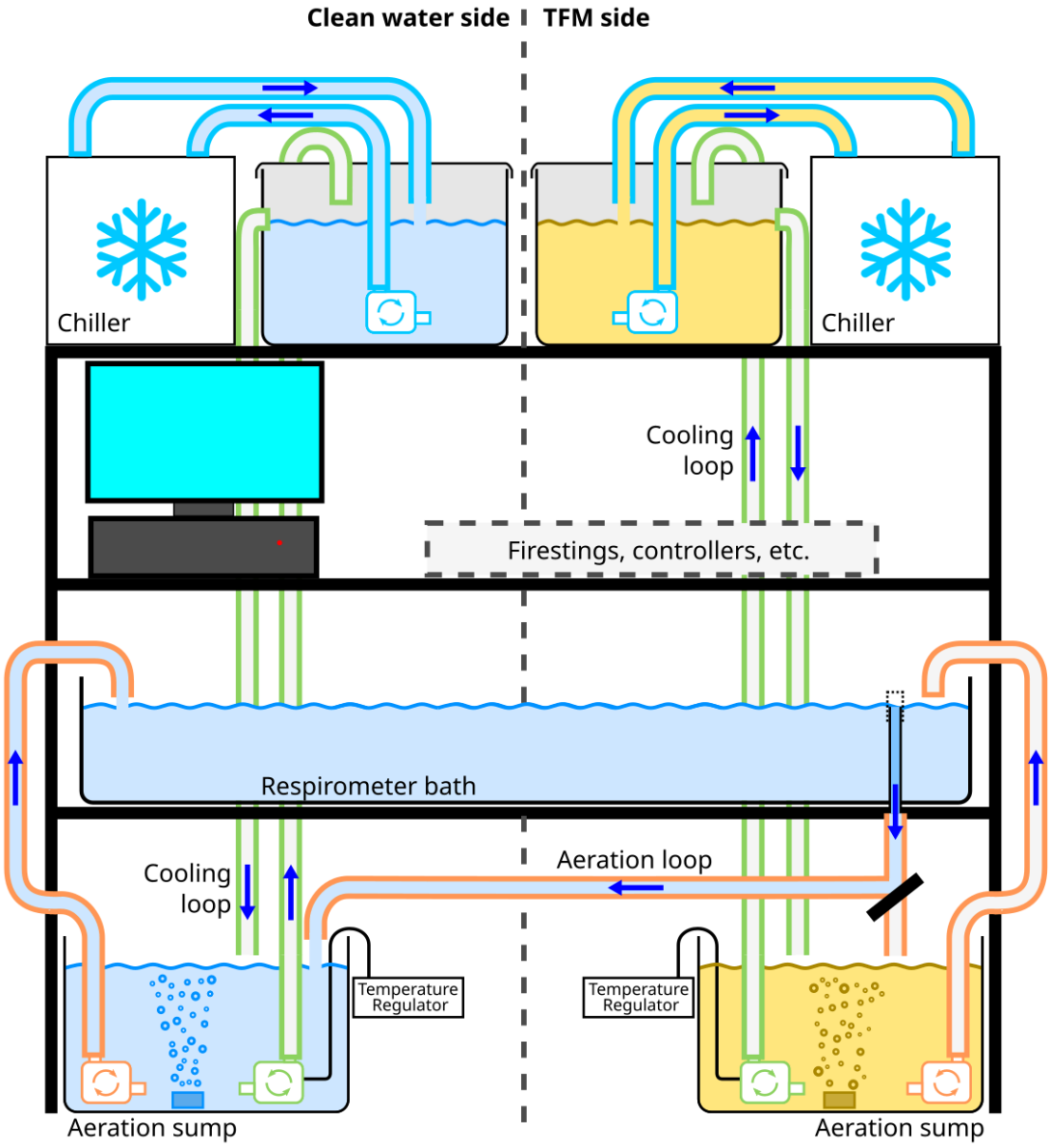
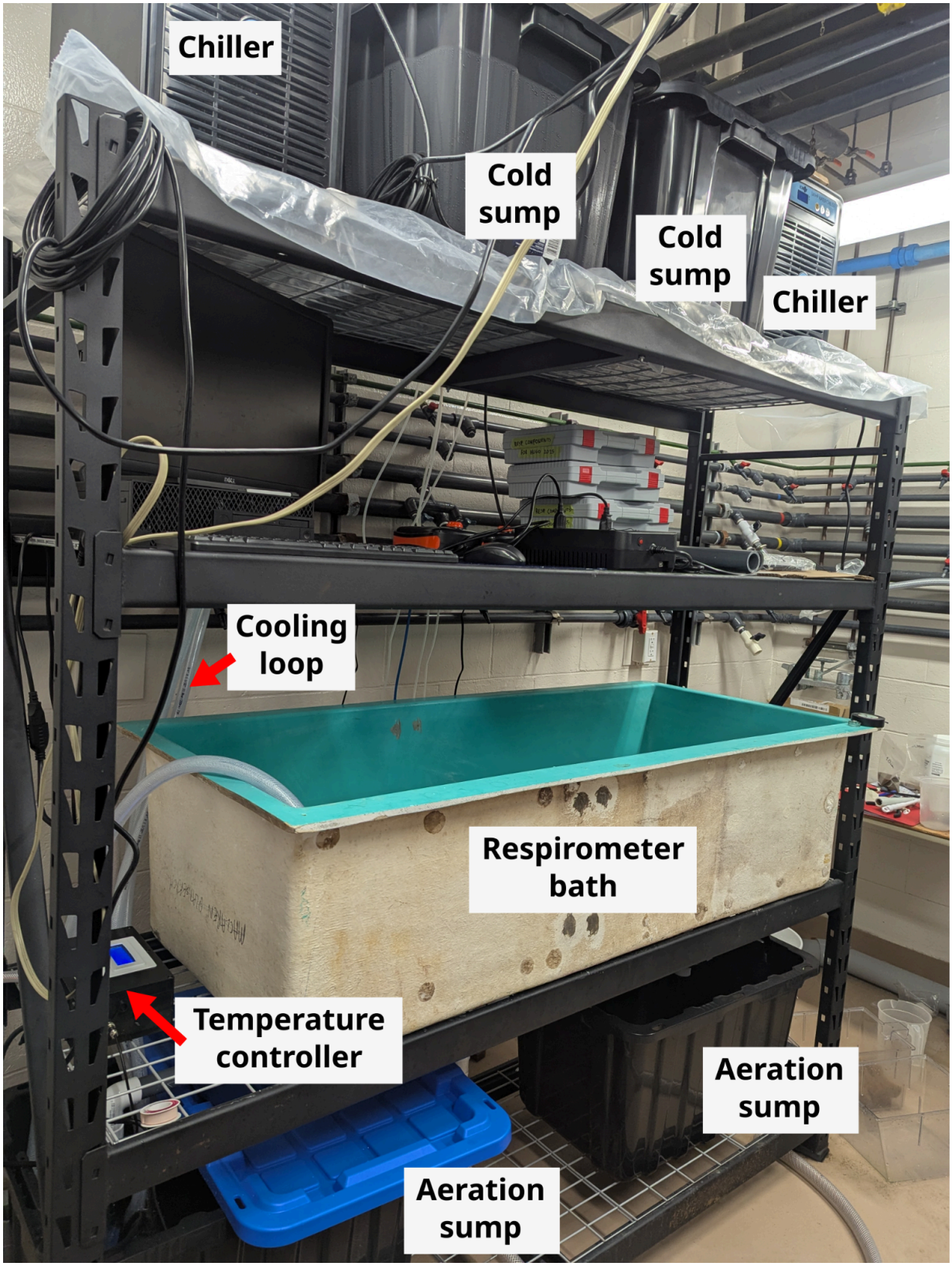


**Photo of the wet table during assembly:**

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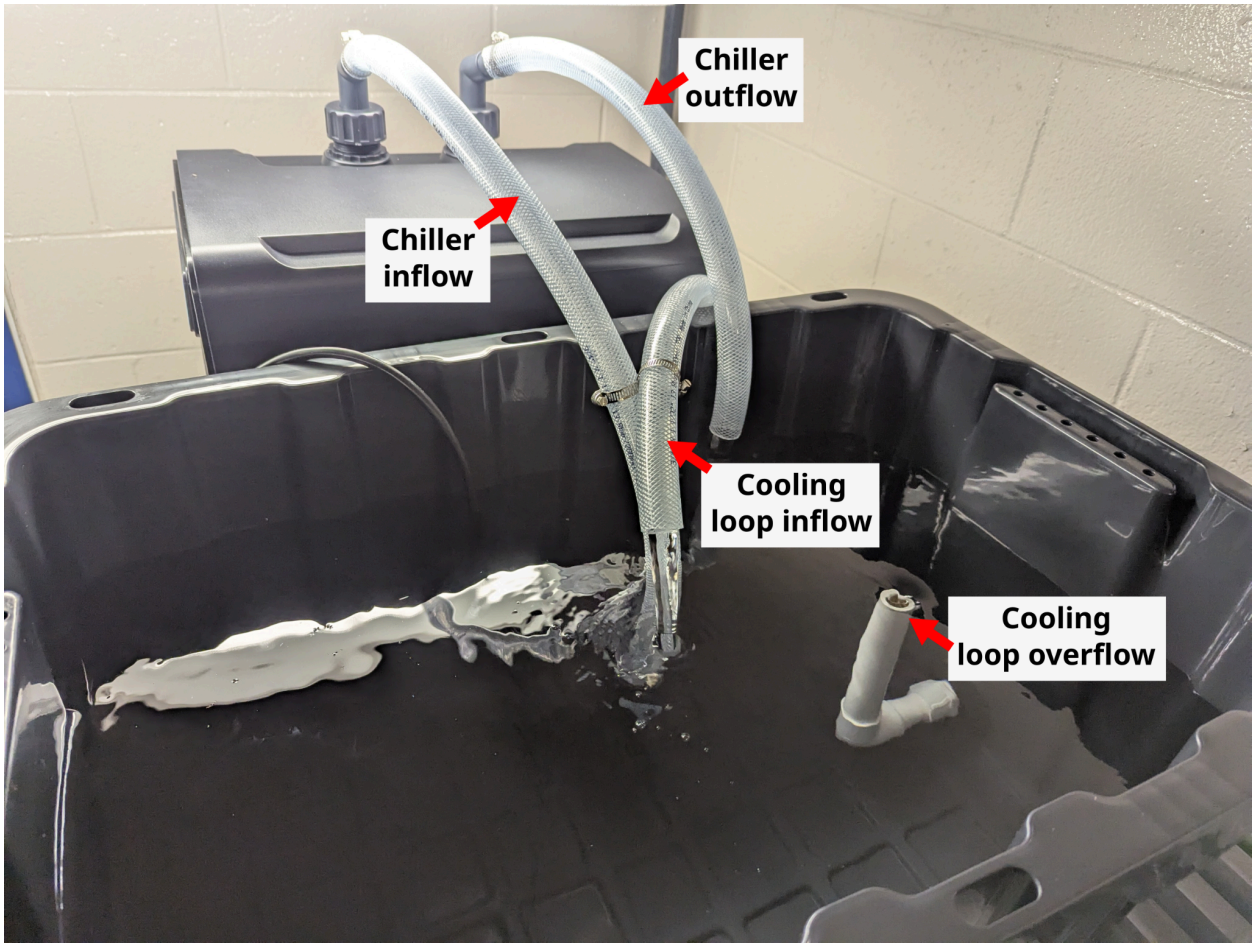
**Close up of the clean water cooling sump:**

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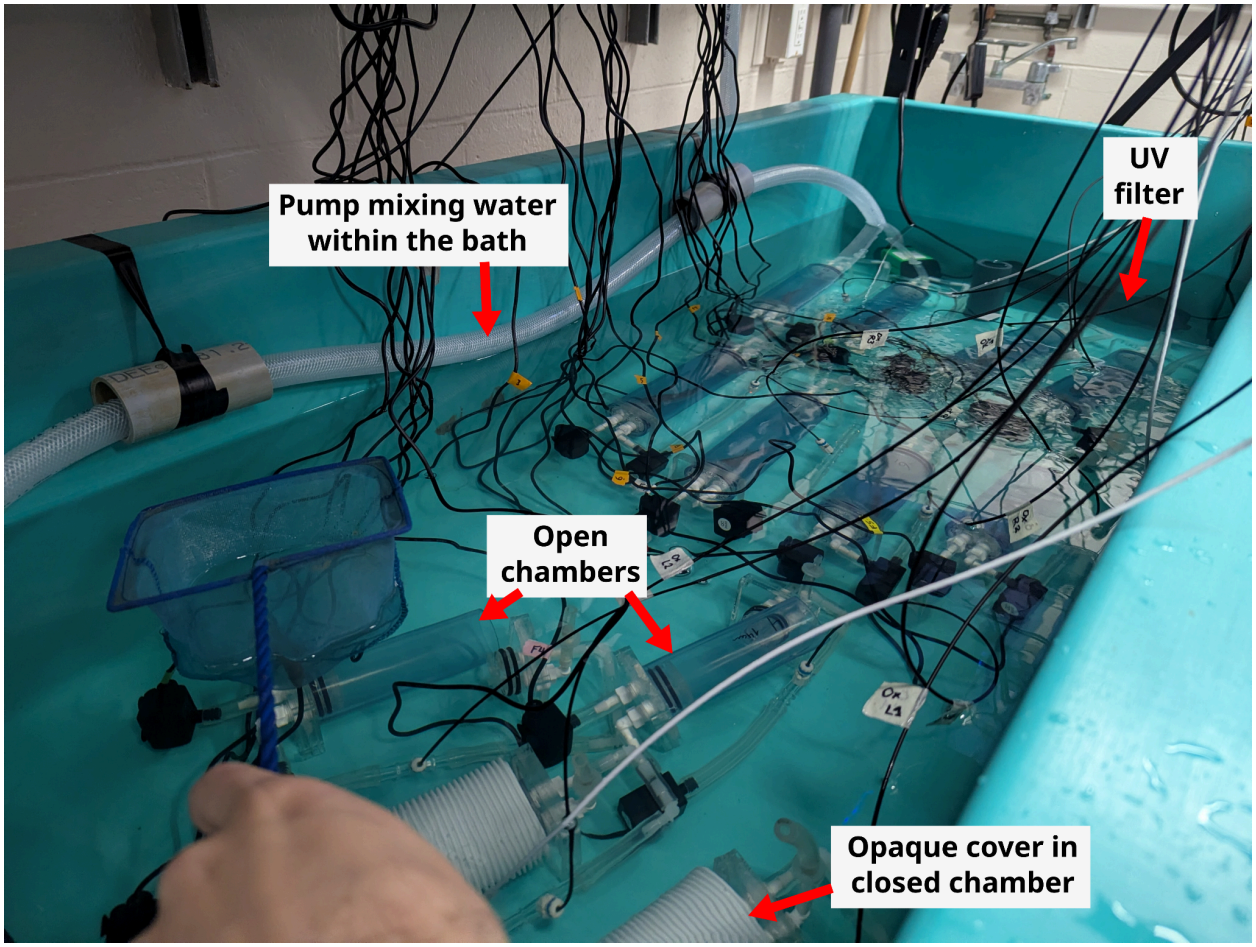
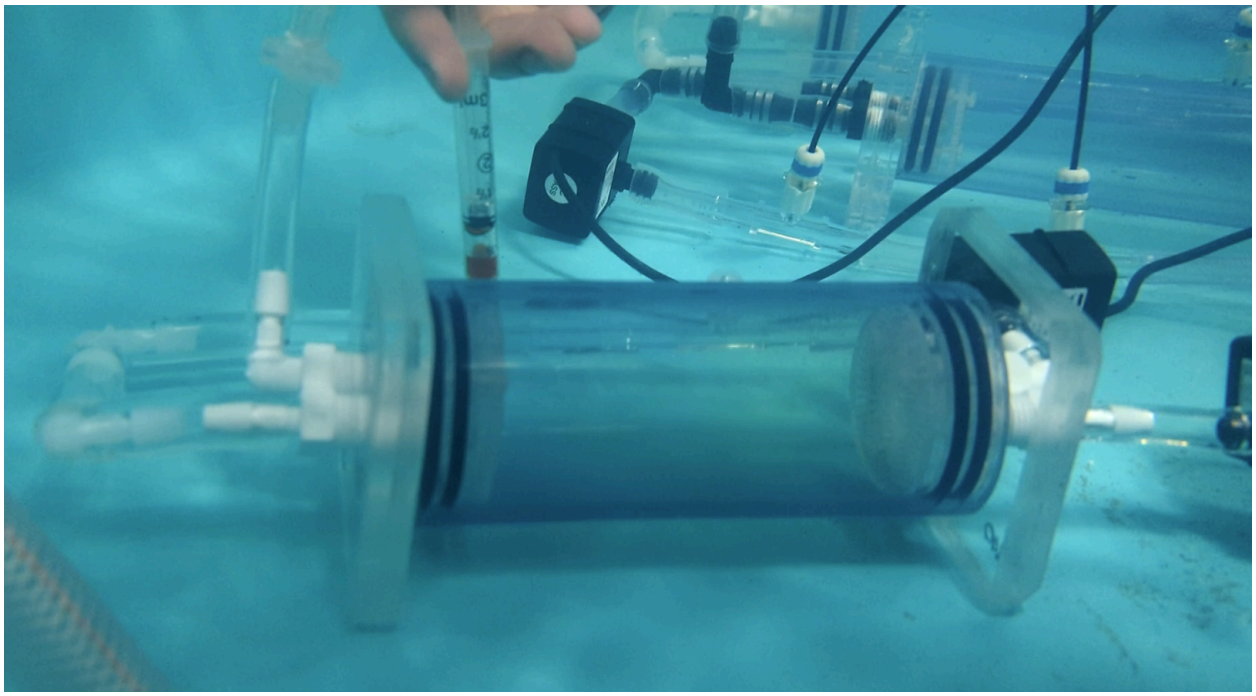
**Respirometer bath with clean water (before trial start):**

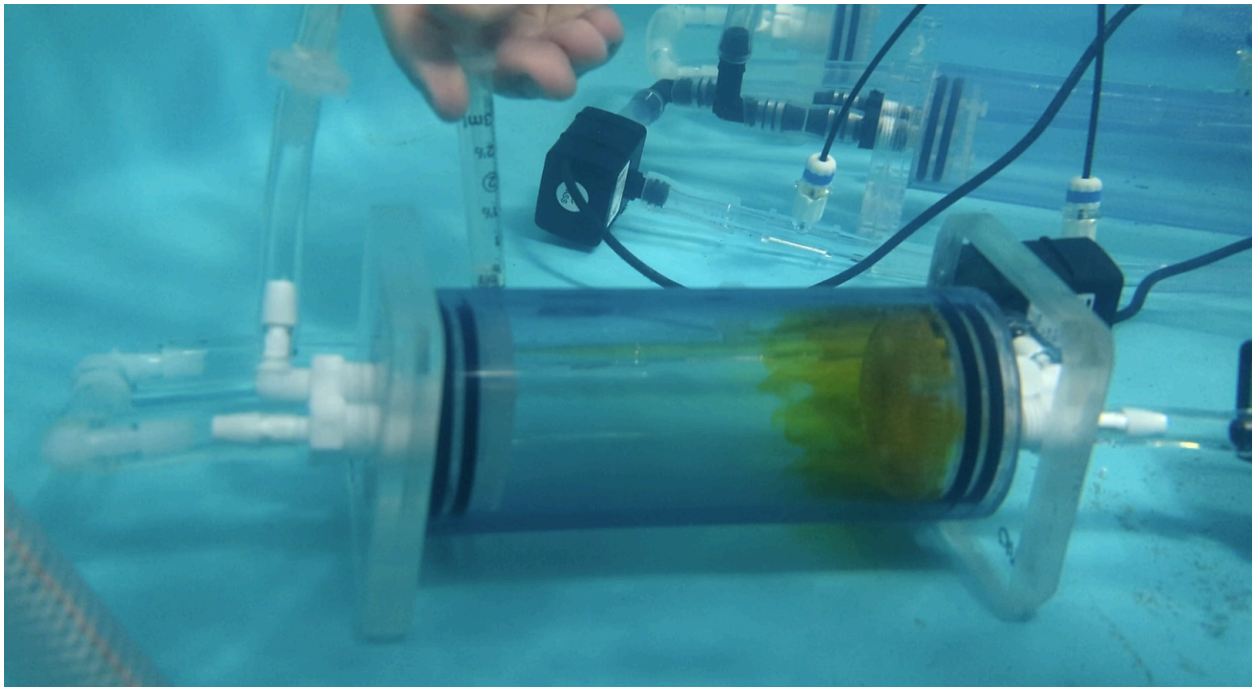



UV filters were turned off during the TFM exposure periods.

**Dye test sequence on one chamber:**

20s video available in the paper's Zenodo repository: www.doi.org/10.5281/zenodo.21877281





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